Multi-User Security of LightMAC and LightMAC_Plus

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Abstract. LightMAC is one of the ISO/IEC standardized message authentication codes that provably achieves security roughly in the order of $O(q^2/2^n)$, where q is the total number of queries and n is the block size of the underlying block cipher. In a subsequent work, Naito proposed a beyond-birthday-bound variant of the LightMAC construction, dubbed LightMAC_Plus, and demonstrated that it achieves $2n/3$-bit PRF security. Later in EUROCRYPT'20, Kim et al. improved the security bound of LighMAC_Plus from $2n/3$ bits to $3n/4$ bits. However, all these security results have been proven in the single-user setting, where we assume that the adversary has access to a single instance of the construction. In this paper, we investigate, for the first time, the security of the LightMAC and the LightMAC_Plus constructions in the context of the multi-user setting, where we assume that the adversary has access to more than one instance of the construction. In particular, we have shown that LightMAC offers $O(qq_{\max}\ell k/2^n + up/2^k)$ multi-user PRF security, where q denotes the total number of construction queries, p denotes the total number of offline primitive queries, $q_{\max}$ denotes the maximum number of queries per user, u denotes the total number of users and ℓ denotes the maximum number of message blocks in a query. We have also shown that LightMAC_Plus maintains security up to approximately $2^{2n/3}$ construction queries and $2^{2k/3}$ ideal-cipher queries in the ideal-cipher model, where n denotes the block size and k denotes the key size of the block cipher.

Keywords: LightMAC, LightMAC_Plus, Multi-user Security, Mirror Theory, Beyond Birthday Bound.

1 Introduction

Over the past few decades, research interest in lightweight cryptography has experienced remarkable growth within the cryptographic community. Lightweight cryptography is designed to secure communications in resource-constrained environments. With the advent of the Internet of Things (IoT), lightweight cryptography has gained significant momentum in the last decade. As a consequence of that, the cryptographic community started to realize standardizing the lightweight cryptographic algorithms through various competitions and projects, most notably the CAESAR competition [CAE14], the NIST lightweight cryptography standardization project [NIS18], and the ISO/IEC standardization [2719]. In this regard, ISO/IEC 29192-6:2019 standard [2719] specifies three message authentication code (or MAC) algorithms for lightweight applications; LightMAC [LPTY16], Tsudik's keymode [Tsu92], and Chaskey-12 [Mou15].

1.1 LightMAC Construction

In FSE'16 [LPTY16], Luykx et al. have proposed $\mathsf{LightMAC}$, which has been standardized by the ISO/IEC standardization process. $\mathsf{LightMAC}$ is a block cipher-based PRF that operates in parallel mode. For an n-bit block cipher E instantiated with two independently sampled keys K_1, K_2, and with a global counter size s, the $\mathsf{LightMAC}$ function is defined as follows:

$$\mathsf{LightMAC}_{\mathsf{E}_{K_1,K_2}}(M) = \mathsf{E}_{K_2}\Bigg(\overbrace{\sum_{\alpha=1}^{\ell-1} \underbrace{\mathsf{E}_{K_1}(\langle\alpha\rangle_s \| M[\alpha])}_{V[\alpha]} \oplus (M[\ell]\|10^{\star})}^{\mathsf{LightMAC\text{-}Hash}_{\mathsf{E}_{K_1}}}\Bigg),$$

where $(M[1], \ldots, M[\ell])$ denotes the $(n-s)$-bit parsing of the input message M such that the bit length of $M[i]$, for $1 \leq i \leq \ell - 1$, is exactly $n - s$ and that of $M[\ell]$ is at most $n - s$. $M[\ell]\|10^*$ denotes the concatenation of $M[\ell]$, which is of length at most $n - s$ bits, with bit 1 followed by a minimum number of 0's so that the length of the resulting bit string $M[\ell]\|10^*$ becomes exactly n. $\langle\alpha\rangle_s$ denotes the s-bit binary representation of counter α. A pictorial description of the construction is shown in Fig. 1. The rate of this construction is $1 - s/n$ [1].

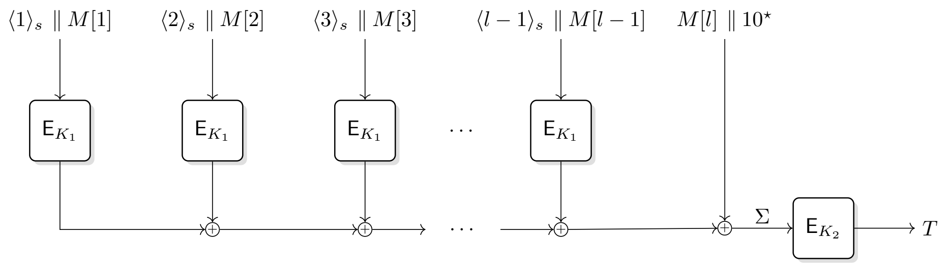

Figure 1: The LightMAC Construction. Here, $M[\ell]\|10^*$ denotes the concatenation of $M[\ell]$, which is of length at most $n - s$ bits, with a lone bit 1 followed by the minimum number of 0's so that the length of the resulting string becomes exactly n.

The design of $\mathsf{LightMAC}$ is simple as it minimizes all auxiliary operations (by just using XOR) other than having the block cipher calls, which allows a low overhead cost, and hence offers a compact implementation. Moreover, due to the inherent parallelism in the design, it outperforms all the other popular sequential MAC constructions in terms of throughput in the parallel computing infrastructure. One of the other features of $\mathsf{LightMAC}$ is its length-independent security bound, which is not present in its contender candidates [BR02, IK03, BKR00, BR00]. However, some variants of PMAC, e.g., PMAC-with-parity [Yas12], $\mathsf{PMAC3}$ [Nai19] etc., achieve message length independent security bound for a wide range of $\ell_{\max}$, they come at the cost of a significant increase in the design complexity.

Related Works on $\mathsf{LightMAC}$: Two simple variants of the $\mathsf{LightMAC}$ construction have been proposed in [SWG21], dubbed as $\mathsf{LedMAC1}$ and $\mathsf{LedMAC2}$. The former one avoids the unnecessary padding of $\mathsf{LightMAC}$ that allows messages up to length $(n-s)2^s + s - 1$ bits without degrading the security bound, whereas the later one is a single-keyed variant of $\mathsf{LedMAC1}$, achieving a similar level of security ($q\sigma/2^n$, where σ denotes the total number of message blocks across all q queries) as that of $\mathsf{LightMAC}$. In [CJN21], Chattopadhyay et al. proposed a single-keyed variant of $\mathsf{LightMAC}$ construction dubbed as $\mathsf{1k\text{-}LightMAC}$, and

[1] the rate of construction is determined by the ratio of the total number of n-bit message blocks in a message M to the total number of primitive calls with block size n required to process the message

showed that it achieves a security bound of $O(q^2/2^n)$ while restricting the query length up to $(n-s)\min\{2^{n/4}, 2^s\}$ bits. A domain-separated variant of 1k-LightMAC construction, called LightMAC-ds has also been proposed with a similar security bound as that of LightMAC with maximum message length up to $(n-s)2^{s-1}$ bits.

1.2 LightMAC_Plus: BBB Secured Variant of LightMAC

In ASIACRYPT'17, Naito [Nai17] proposed a beyond birthday bound secure variant of LightMAC construction called LightMAC_Plus. Similar to the LightMAC construction, LightMAC_Plus is a block cipher based PRF that operates in parallel mode, i.e., for an n-bit block cipher E instantiated with three independently sampled keys K_1, K_2, K_3, and with a global counter size s, the LightMAC_Plus function is defined as follows:

$$\mathsf{LightMAC\_Plus}_{\mathsf{E}_{K_1,K_2,K_3}}(M) = \mathsf{E}_{K_2}(\Sigma) \oplus \mathsf{E}_{K_3}(\Theta),$$

where

$$\Sigma \;:=\; \Bigg(\sum_{\alpha=1}^{\ell} \underbrace{\mathsf{E}_{K_1}(\langle\alpha\rangle_s \| M'[\alpha])}_{V[\alpha]}\Bigg),\ \Theta := \Bigg(\sum_{\alpha=1}^{\ell} 2^{\ell-\alpha}\,\underbrace{\mathsf{E}_{K_1}(\langle\alpha\rangle_s \| M'[\alpha])}_{V[\alpha]}\Bigg).$$

We would like to mention that $(M'[1], \ldots, M'[\ell])$ denotes the $(n-s)$-bit parsing of $M' = M\|10^*$, where, M' is derived from M by appending a bit 1 followed by a minimum number of $0's$ so that the length of the resulting message becomes the multiple of $(n-s)$. As before, $\langle\alpha\rangle_s$ denotes the s-bit binary representation of counter α. A pictorial description of the construction is shown in Fig. 2.

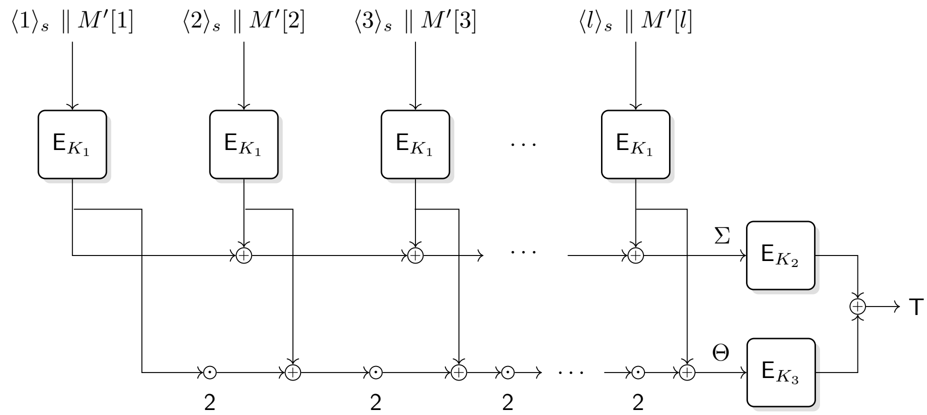

Figure 2: The LightMAC_Plus Construction

Naito has shown that LightMAC_Plus provably achieves a message length independent PRF security of up to $2^{2n/3}$ queries, which has been improved later in [KLL20] from $2n/3$ bits to $3n/4$ bits.

Related Works on LightMAC_Plus: Naito [Nai17] proposed LightMAC_Plus2 that provides a higher security bound than LightMAC_Plus but comes at the increased number of block cipher calls. In [Nai18], Naito has improved the bound of LightMAC_Plus from $q^3/2^{2n}$ to $q_t^2 q_v/2^{2n}$, where q_t is the number of tagging queries and q_v is the number of verification queries. This security bound implies that LightMAC_Plus is secure up to 2^n tagging queries if the number of verification queries is 1. Later, Leurent et al. [LNS18] have shown a forging attack on the construction that achieves a constant success probability when the number of tagging queries is $2^{3n/4}$ and the number of verification queries is 1. In [KLL20], Kim et al. have shown an improved security bound of LightMAC_Plus construction from $2n/3$-bits to $3n/4$-bits (ignoring the maximum message length), which establishes the tightness of the bound due to the result of [LNS18]. Subsequently, several results [DDNP18, DDK23, Nai18, Son21, DDM24] have studied the security of different variants of the LightMAC_Plus construction.

1.3 Multi User Security

The security analysis for all the above-mentioned constructions has been conducted in a single-user setting, where the adversary has access to the keyed construction for a single unknown randomly sampled key. This setup, known as the *single-user security model*, involves the adversary targeting one specific machine, in which the cryptographic algorithm is deployed, to compromise its security. However, in practice, cryptographic algorithms are usually deployed across multiple machines. For instance, AES-GCM [MV04, M.D07] is now widely used in the TLS protocol to protect web traffic and is currently used by billions of users daily. In such multi-user scenarios, the adversary's goal is to compromise the security of at least one user. This approach, known as the *multi-user security model*, considers the adversary's success as a combination of single-user successes.

The notion of multi-user (mu) security was introduced by Biham [Bih02] in symmetric cryptanalysis and by Bellare, Boldyreva, and Micali [BBM00] in the context of public-key encryption. Security analysis of symmetric ciphers in the multi-user setting is quite different from that of in the single-user setting. For example, in the single-key setup, attacks on block ciphers like AES don't improve with more data complexity. However, in the multi-key environment, they do, as observed by Biham [Bih02] and refined (time-memory-data trade-off) by Biryukov et al. [BMS05]. This highlights how it's easier to recover a block cipher key from a large group than to target a specific one. This principle extends to the study of multi-user security of MACs by Chatterjee et al. [CMS11]. Bellare and Tackmann [BT16a] formalized a multi-user secure authenticated encryption scheme and analyzed countermeasures against multi-key attacks in TLS 1.3.

As evident from [BBT16, BT16a, BHT18, HT16, HT17, LMP17, ML15], it is a challenging problem to study the security degradation of cryptographic primitives with the number of users, even when their security in the single-user setting is well understood. Research on the multi-user security of MACs is relatively limited in the literature, with notable works by Chatterjee et al. [CMS11], Morgan et al. [MPS20], Bellare et al. [BT16b], and two recent works of Shen et al. [SWGW21] and Datta et al. [DDNT23]. The first two employ a generic reduction approach for MACs, multiplying the single-user security by the number of users, while the latter employs a dedicated analysis method for the DbHtS [DDNP18] construction. However, these results cannot be directly applied to the multi-user security of LightMAC and LightMAC_Plus. Therefore, to address the applicability of the LightMAC and LightMAC_Plus constructions in practical setting, we ask the following:

> *How does the security of* LightMAC *and* LightMAC_Plus *degrade in the multi-user setting ?*

1.4 Our Contribution

In this paper, we address the above question and study the multi-user security of LightMAC and LightMAC_Plus constructions.

I. Multi-user Security of LightMAC. We first derive the multi-user PRF security bound of the LightMAC construction based on the assumption that the underlying block cipher of the LightMAC construction is an ideal cipher. In particular, we prove that if the block cipher E is an ideal cipher with key space $\{0,1\}^k$ and the message space $\{0,1\}^n$, then the construction is secure against u users that make a total of q construction queries and p offline primitive queries such that each user is limited to make at most $q_{\max}$ queries and the number of message blocks in a query is at most ℓ, with the following advantage:

$$\begin{aligned}\mathbf{Adv}^{\text{mu-PRF}}_{\mathsf{LightMAC[E]}}(u,q,q_{\max},p,\ell) \quad &\leq \quad \frac{2u^2}{2^k}+\frac{2qq_{\max}}{2^n}+\frac{28p}{2^k}+\min\{\frac{up}{2^k},\frac{q\ell k}{2^n}\}\\ &+\min\{\frac{up}{2^k},\frac{qq_{\max}\ell k}{2^n},\frac{p\sqrt{q}}{2^k}+\frac{q^{3/2}k\ell}{2^n}\}.\end{aligned}$$

By instantiating the construction with AES-128-256, we obtain ℓ-free security bound. Moreover, by limiting the maximum number of queries per user to 2^{32}, we obtain beyond birthday bound security in terms of q, provided the number of users is up to 2^{64}. While instantiating the construction with AES-128-128, we observed that the security bound of the construction is affected by the maximum number of message blocks. However, suppose we limit the number of users to 2^{15}. In that case, we obtain an ℓ-free security bound with the maximum number of queries per user limited to 2^{56} and the total number of queries reaching up to 2^{72}. We refer the readers to Section 3.2 and Table 1 for a complete view of different cases of the security bound of LightMAC construction.

II. Multi-user Security of LightMAC_Plus. For LightMAC_Plus, we show that the construction is secure in the multi-user setting against all adversaries that make a total of at most $2^{2n/3}$ construction queries and up to $2^{2k/3}$ primitive queries to the underlying block cipher, where we assume that E is an ideal cipher with key space $\{0,1\}^k$ and message space $\{0,1\}^n$. More generally, if we consider u users making a total of q construction queries and p primitive queries, such that each user makes at most $q_{\max}$ queries and each query consists of at most ℓ message blocks, then the following multi-user PRF advantage bound holds:

$$\begin{aligned}\mathbf{Adv}^{\text{mu-PRF}}_{\mathsf{LightMAC\_Plus[E]}}(u,q,q_{\max},p,\ell) \quad &\leq \tfrac{2u}{2^k}+\tfrac{u^3}{6\cdot 2^{2k}}+\tfrac{2up^2}{2^{2k}}+\tfrac{6q^2\ell}{2^{n+k}}+\tfrac{5qq_{\max}^2}{2^{2n}}+\tfrac{2p\sqrt{q}}{2^k}\\ &+\tfrac{2q^{3/2}k\ell}{2^n}+\tfrac{2q\sqrt{q_{\max}}}{2^n}+\min\left\{\tfrac{2up}{2^k},\tfrac{2q\ell k}{2^n}\right\}+\tfrac{52p}{2^k}\\ &+\tfrac{40p^2q^2}{2^{3n}}+\tfrac{320(p+q)q^3}{2^{3n}}+\tfrac{160p^2q}{2^{2n}}+\tfrac{1280(p+q)q^2}{2^{2n}},\end{aligned}$$

valid under the conditions $4q < 2^n$, $6p < 2^k$ and $\ell \leq \min\{2^{n-1}, 2^s\}$. Moreover, if we assume that $k \geq 4n/3$, then we obtain an ℓ-free bound, provided $\ell \leq 2^{n/3}/k$, with the following advantage:

$$\begin{aligned}\mathbf{Adv}^{\text{mu-PRF}}_{\mathsf{LightMAC\_Plus[E]}}(u,q,q_{\max},p) \quad &\leq \tfrac{2u}{2^k}+\tfrac{u^3}{6\cdot 2^{2k}}+\tfrac{2up^2}{2^{2k}}+\tfrac{5q^2}{n2^{2n}}+\tfrac{5qq_{\max}^2}{2^{2n}}+\tfrac{2p\sqrt{q}}{2^k}\\ &+\tfrac{2q^{3/2}}{2^n}+\tfrac{2q\sqrt{q_{\max}}}{2^n}+\min\left\{\tfrac{2up}{2^k},\tfrac{2q}{2^{2n/3}}\right\}+\tfrac{13p}{2^{3k/4}}\\ &+\tfrac{40p^2q^2}{2^{3n}}+\tfrac{320(p+q)q^3}{2^{3n}}+\tfrac{160p^2q}{2^{2n}}+\tfrac{1280(p+q)q^2}{2^{2n}}.\end{aligned}$$

We emphasize that, unlike the single-user case, the proposed multi-user security bound of LightMAC_Plus is not ℓ-free in general. Furthermore, the security for primitive queries is only $2k/3$ bits, which is smaller than the key size. On the other hand, in the standard model, the multi-user security is $k - \log_2 u$ bits. Moreover, LightMAC_Plus achieves $3n/4$-bit security in the standard model, which is stronger than the current bound. We believe that improving the multi-user bound in the ideal-cipher model from $2n/3$ bits to $3n/4$ bits is a hard problem.

Remark 1. The single-user security bound for the LightMAC and LightMAC_Plus constructions is established in the standard model. The overall bound includes an information-theoretic term that is ℓ-free, and a computational term that relies on the security of the underlying block cipher, which involves an upper limit on the number of queried blocks. Our bounds give a clearer picture of what happens under the hood. However, we would like to clarify that a proof in the ideal cipher model, where we treat E as an ideal block cipher, uniformly sampled from the set of all block ciphers, does not guarantee the security of the LightMAC and LightMAC_Plus constructions for any instantiation of the construction with an actual block cipher.

1.5 How Does Our Result Differ From [SWGW21, DDNT23]?

In [SWGW21], Shen et al. have analyzed the multi-user security of the DbHtS construction and shown it to be secure roughly up to $2^{2n/3}$ construction queries and 2^k ideal-cipher queries. They have applied their generic security result to bound the multi-user security of different DbHtS type constructions that include SUM-ECBC [Yas10], PMAC_Plus [Yas11], 3kf9 [ZWSW12], LightMAC_Plus [Nai17], etc. However, as pointed out in [DDNT23], this application was not appropriate as the security analysis assumed the underlying hash function is a keyed function (which may not be a block cipher-based hash function), and thus the security proof has not analyzed the properties of the hash function in the ideal-cipher model. As a result, when the generic security bound was applied to bound the multi-user security of block cipher-based DbHtS constructions, the authors failed to consider the fact that the block cipher used in the hash function is an ideal cipher, where the adversary is allowed to make ideal-cipher queries. Therefore, to prove the multi-user security of the above block cipher-based DbHtS constructions, one needs to do a dedicated security analysis without resorting to any generic security result.

In [DDNT23], Datta et al. have shown a tight multi-user security bound on the two-keyed DbHtS construction with $2^{3n/4}$ queries and 2^k ideal-cipher queries. They have also instantiated the underlying double block hash function with an algebraic keyed hash function and shown its $3n/4$-bit multi-user security. Extending their analysis to a block cipher-based construction was posed as an open problem in [DDNT23].

Our result is different from the previous works in the sense that we have established the multi-user security bound on two particular constructions LightMAC and LightMAC_Plus using a dedicated security analysis. We have been able to show that the multi-user security bound of LightMAC does not degrade much compared to its security bound in the single-user setting. On the other hand, we have been able to show that LightMAC_Plus construction achieves $2n/3$-bit multi-user security bound when the number of ideal-cipher queries is restricted up to $2^{2k/3}$. We would like to mention that Naito [Nai24] has recently shown multi-user security bounds of HtE and HtXE type constructions based on the assumption that the underlying block cipher is an ideal cipher and the underlying hash function is an almost-xor-universal hash function in the ideal cipher model. Finally, they have applied their result to bound the multi-user security of CBC, EMAC, XCBC, and TMAC.

2 Preliminaries

General Notations: For $q \in \mathbb{N}$, we write $[q]$ to denote the set $\{1, \ldots, q\}$. For a natural number n, $\{0,1\}^n$ denotes the set of all binary strings of length n, and $\{0,1\}^*$ denotes the set of all binary strings of arbitrary length. For a natural number n, we call the elements of $\{0,1\}^n$ *block*. For any binary string $x \in \{0,1\}^*$, $|x|$ denotes the length of x, i.e., the

number of bits in x. For $x, y \in \{0,1\}^n$, we write $z = x \oplus y$ to denote the xor of x and y. For two binary strings $x, y \in \{0,1\}^*$, we write $x \| y$ to denote the concatenation of x followed by y. For a natural number n and $x \in \{0,1\}^*$, we write $(x_1, x_2, \ldots, x_{l-1}, x_l) \xleftarrow{n} x$ to denote the n-bit parsing of x, where $|x_i| = n$ for all $i \in [l-1]$ and $0 \leq |x_l| \leq n-1$. Typically, a tuple or a vector x over $\{0,1\}^n$ is denoted as $(x_i)_{i \in \mathcal{I}}$, where $\mathcal{I}$ is called the *index set* and each $x_i \in \{0,1\}^n$. However, when there are two index sets $\mathcal{I}, \mathcal{J}$, then we denote a tuple x over $\{0,1\}^n$ as $(x^i_j)_{i \in \mathcal{I}, j \in \mathcal{J}}$, where each x^i_j is an element of $\{0,1\}^n$. Extending it one step further, we denote a tuple x over $\{0,1\}^n$, when the number of index sets is three, i.e., $\mathcal{I}, \mathcal{J}$ and $\mathcal{B}$, as $x = (x^i_j[\alpha])_{i \in \mathcal{I}, j \in \mathcal{J}, \alpha \in \mathcal{B}}$.
For two positive integers i, s such that $i < 2^s$, we write $\langle i \rangle_s$ to denote the s-bit binary representation of integer i. We write $x \leftarrow y$ to denote the assignment of the variable y into x. $\mathsf{X} \leftarrow\!\!\$\ \{0,1\}^n$ denotes that X is sampled uniformly at random from $\{0,1\}^n$. For a tuple of random variables $(X_1, \ldots, X_q)$, we write $(X_1, \ldots, X_q) \xleftarrow{\text{wor}} \{0,1\}^n$ to denote that each X_i is sampled uniformly from $\{0,1\}^n \setminus \{X_1, \ldots, X_{i-1}\}$, i.e., $X_i \leftarrow\!\!\$\ \{0,1\}^n \setminus \{X_1, \ldots, X_{i-1}\}$. We denote the set of all functions from $\mathcal{X}$ to $\{0,1\}^n$ as $\mathsf{Func}_{\mathcal{X}}$. Sometimes, we omit the set $\mathcal{X}$ from $\mathsf{Func}_{\mathcal{X}}$ and write Func when the domain is clear from the context. The set of all permutations over $\{0,1\}^n$ is denoted as Perm. We write $\mathbf{P}(n, r)$ to denote $n(n-1)\ldots(n-r+1)$, where $(n)_0 = 1$ by convention, which represents the number of ways to arrange r different items from a set of n different items.

2.1 Pseudorandom Function and Pseudorandom Permutation

Let $\mathsf{F} : \{0,1\}^k \times \mathcal{X} \to \{0,1\}^n$ be a family of keyed functions from $\mathcal{X}$ to $\{0,1\}^n$. We define the pseudorandom function (prf) advantage of F with respect to a distinguisher A as follows:

$$\mathbf{Adv}^{\mathrm{prf}}_{\mathsf{F}}(\mathsf{A}) \stackrel{\Delta}{=} \left|\mathsf{Pr}[K \leftarrow \{0,1\}^k : \mathsf{A}^{\mathsf{F}_K} = 1] - \mathsf{Pr}[\mathsf{R} \leftarrow \mathsf{Func} : \mathsf{A}^{\mathsf{R}} = 1]\right|.$$

We say that F is $(q, \ell, \sigma, \mathtt{t}, \epsilon)$ secure if the maximum pseudorandom function advantage of F is ϵ, where the maximum is taken over all distinguishers A that make q queries to its oracle such that the total number of message blocks queried across all q queries is σ with ℓ being the maximum number of message blocks among all q queries, and the adversary runs for time at most $\mathtt{t}$, i.e.,

$$\mathbf{Adv}^{\mathrm{prf}}_{\mathsf{F}}(q, \ell, \sigma, \mathtt{t}) \stackrel{\Delta}{=} \max_{\mathsf{A} \in \mathcal{C}} \mathbf{Adv}^{\mathrm{prf}}_{\mathsf{F}}(\mathsf{A}),$$

where $\mathcal{C}$ is the class of all distinguishers A that makes at most q queries such that the total number of message blocks queried across all q queries is σ with ℓ being the maximum number of message blocks among all q queries with run time at most $\mathtt{t}$. Similarly, we define the notion of pseudorandom permutation advantage as follows:
Let $n, k \in \mathbb{N}$ be two natural numbers. Let $\mathsf{E} : \{0,1\}^k \times \{0,1\}^n \to \{0,1\}^n$ be a family of keyed functions from $\{0,1\}^n$ to $\{0,1\}^n$ such that for $K \in \{0,1\}^k$, the function $\mathsf{E}_K : \{0,1\}^n \to \{0,1\}^n$ is bijective. We define the pseudorandom permutation (prp) advantage of E with respect to a distinguisher A as follows:

$$\mathbf{Adv}^{\mathrm{prp}}_{\mathsf{E}}(\mathsf{A}) \stackrel{\Delta}{=} \left|\mathsf{Pr}[K \leftarrow \{0,1\}^k : \mathsf{A}^{\mathsf{E}_K} = 1] - \mathsf{Pr}[\mathsf{P} \leftarrow \mathsf{Perm} : \mathsf{A}^{\mathsf{P}} = 1]\right|.$$

We call the family of keyed functions E a *block cipher* with the *block size* n-bits and the *key size* k-bits. We say that E is $(q, \mathtt{t}, \epsilon)$ secure if the maximum pesudorandom permutation advantage of E is ϵ, where the maximum is taken over all distinguishers A that make q queries to the respective oracle and run for time at most $\mathtt{t}$, i.e.,

$$\mathbf{Adv}^{\mathrm{prp}}_{\mathsf{E}}(q, \mathtt{t}) \stackrel{\Delta}{=} \max_{\mathsf{A} \in \mathcal{C}} \mathbf{Adv}^{\mathrm{prp}}_{\mathsf{E}}(\mathsf{A}),$$

where $\mathcal{C}$ is the class of all distinguishers A that makes at most q queries with run time at most $\mathtt{t}$. We write $\mathsf{BC}(\mathcal{K}, \{0,1\}^n)$ to denote the set of all block ciphers with key space $\mathcal{K}$ and block size n bits.

2.2 Expectation Method and H-Coefficient Technique

Hoang and Tessaro [HT16] introduced the Expectation Method to derive a tight multi-user security bound of the key-alternating cipher. Subsequently, this technique has been used for bounding the distinguishing advantage of various cryptographic constructions [HT17, BHT18,DNT19]. The Expectation Method is a generalization of the H-Coefficient technique developed by Patarin [Pat08], which serves as a "systematic" tool to upper bound the distinguishing advantage of any deterministic and computationally unbounded distinguisher A in distinguishing the real oracle $\mathcal{O}_1$ (construction of interest) from the ideal oracle $\mathcal{O}_0$ (idealized version). The collection of all the queries and responses that A made and received to and from the oracle is called the *transcript* of A, denoted as τ. Sometimes, we allow the oracle to release more internal information to A only after A completes all its queries and responses, but before it outputs its decision bit. Note that revealing extra information will only increase the advantage of the distinguisher.
Let X_{re} and X_{id} denote the transcript random variable induced by the interaction of A with the real oracle and the ideal oracle respectively. The probability of realizing a transcript τ in the ideal oracle (i.e., $\mathsf{Pr}[\mathsf{X}_{\text{id}} = \tau]$) is called the *ideal interpolation probability*. Similarly, one can define the *real interpolation probability*. A transcript τ is said to be *attainable* with respect to A if the ideal interpolation probability is non-zero (i.e., $\mathsf{Pr}[\mathsf{X}_{\text{id}} = \tau] > 0$). We denote the set of all attainable transcripts by $\mathcal{V}$. Following these notations, we state the main result of the Expectation Method in Theorem 1. The proof of this theorem can be found in [HT16].

Theorem 1. *Let $\mathcal{V} = \mathsf{GoodT} \sqcup \mathsf{BadT}$ be a partition of the set of attainable transcripts. Let $\Phi : \mathcal{V} \to [0, \infty)$ be a non-negative real valued function. For any attainable good transcript $\tau \in \mathsf{GoodT}$, assume that*

$$\frac{\mathsf{Pr}[\mathsf{X}_{\text{re}} = \tau]}{\mathsf{Pr}[\mathsf{X}_{\text{id}} = \tau]} \geq 1 - \Phi(\tau),$$

and there exists $\epsilon_{\text{bad}} \geq 0$ such that $\mathsf{Pr}[\mathsf{X}_{\text{id}} \in \mathsf{BadT}] \leq \epsilon_{\text{bad}}$. Then,

$$\mathbf{Adv}^{\mathcal{O}_0}_{\mathcal{O}_1}(\mathsf{A}) \leq \mathbf{E}[\Phi(\mathsf{X}_{\text{id}})] + \epsilon_{\text{bad}}. \tag{1}$$

If $\mathcal{O}_0$ is a random function, then Eqn. (1) gives the PRF advantage of A in distinguishing the real construction $\mathcal{O}_1$ from the random function, and in that case, we write

$$\mathbf{Adv}^{\text{PRF}}_{\mathcal{O}_1}(\mathsf{A}) \leq \mathbf{E}[\Phi(\mathsf{X}_{\text{id}})] + \epsilon_{\text{bad}}. \tag{2}$$

We would like to mention that if Φ is a constant function, let $\Phi(\tau) = \epsilon_{\text{ratio}}$, then the Expectation method [HT16] boils down to the H-Coefficient technique [Pat08] and in that case, we have

$$\mathbf{Adv}^{\text{PRF}}_{\mathcal{O}_1}(\mathsf{A}) \leq \epsilon_{\text{ratio}} + \epsilon_{\text{bad}}. \tag{3}$$

2.3 Combinatorial Results

In this section, we state and prove the two results. The first one is from linear algebra, which establishes an upper bound on the probability that a system of equations holds when the variables are wor samples. The second result is about maximizing a function, which will be useful in deriving the security bound of $\mathsf{LightMAC}$ and $\mathsf{LightMAC\_Plus}$ constructions.

Lemma 1. *Let* $(Z_1, \ldots, Z_q) \xleftarrow{\text{wor}} \mathcal{X} \subseteq \{0,1\}^n$ *with* $|\mathcal{X}| = N > q$. *Let* A *be a* $k \times q$ *binary matrix with rank* r. *We denote the column vector* $(Z_1, \ldots, Z_q)^{\mathtt{tr}}$ *as* $\widetilde{\mathsf{Z}}$. *Then, for any* $\widetilde{\mathsf{c}} \in (\{0,1\}^n)^k$, *we have*

$$\Pr[A \cdot \widetilde{\mathsf{Z}} = \widetilde{\mathsf{c}}] \leq \frac{1}{\mathbf{P}(N - q + r, r)}.$$

Proof. The number of ways we can choose the coefficients of the non-basis vectors is at most $\mathbf{P}(N, q - r)$, which uniquely determines the coefficient of the basis vectors. Moreover, the number of ways we can choose q-many variables is $\mathbf{P}(N, q)$. Dividing the former by the latter gives the required probability bound. □

Lemma 2. *Let* q *be a positive non-zero integer and for all* $i \in [q]$, $c_i, h_i \in \mathbb{Z}^+ \cup \{0\}$. *Then the function*

$$c_1 h_1^2 + c_2 h_2^2 + \ldots + c_q h_q^2$$

subject to the constraint

$$c_1 h_1 + c_2 h_2 + \ldots + c_q h_q \leq q,$$

attains the maximum value $q^{3/2}$ *when* $c_1 h_1 = q$, *given that* $h_i \leq \sqrt{q}$ *for all* $i \leq q$ *and* $h_1 \geq h_2 \geq \ldots \geq h_q$.

Proof. Let us consider z_i to denote $c_i h_i$, for all $i \in [q]$. Since, $c_i, h_i \in \mathbb{Z}^+ \cup \{0\}$, it holds that $z_i \in \mathbb{Z}^+ \cup \{0\}$, for all $i \in [q]$. Therefore, the objective function becomes

$$f(z_1, z_2, \ldots, z_q) \stackrel{\Delta}{=} z_1 h_1 + z_2 h_2 + \ldots + z_q h_q,$$

subject to the constraint

$$z_1 + z_2 + \ldots + z_q \leq q.$$

First of all, it is easy to see that the objective function will become maximized when $z_1 + z_2 + \ldots z_q = q$. Therefore, we consider the constraint to be

$$z_1 + z_2 + \ldots z_q = q.$$

Now, our claim is that $v^* \stackrel{\Delta}{=} (q, 0, 0, \ldots, 0)$ is the optimal solution for which the objective function will attain the maximum value. Let us consider some other arbitrary solution $v' \stackrel{\Delta}{=} (q', \alpha_1, \alpha_2, \ldots, \alpha_{q-1})$, where $q' < q$. Then, we have

$$A \stackrel{\Delta}{=} f(v^*) = q h_1, \;\; B \stackrel{\Delta}{=} f(v') = q' h_1 + \alpha_1 h_2 + \alpha_2 h_3 + \ldots + \alpha_{q-1} h_q.$$

Since, v' is a solution, we have $q' = q - (\alpha_1 + \alpha_2 + \ldots + \alpha_{q-1})$. Plugging-in the above equality in B yields

$$f(v') = q h_1 - (h_1 - h_2)\alpha_1 - (h_1 - h_3)\alpha_2 - \ldots - (h_1 - h_q)\alpha_{q-1}.$$

Since, $h_1 \geq h_i$, for all $i \in [2, q]$, it holds that $f(v') \leq f(v^*)$. Therefore, the maximum value obtained, is $q h_1 \leq q^{3/2}$, as $h_1 \leq \sqrt{q}$. □

2.4 Mirror Theory

Consider an undirected edge-labelled bipartite graph $\mathsf{G} = (\mathcal{V}_1 \sqcup \mathcal{V}_2, \mathcal{E}, \mathcal{L})$ with edge labelling function $\mathcal{L} : \mathcal{E} \rightarrow \{0,1\}^n$, where $\mathcal{V}_1 = \{Y_1, \ldots, Y_{s_\ell}\}$ and $\mathcal{V}_2 = \{Z_1, \ldots, Z_{s_{\mathtt{r}}}\}$, such that $s = s_\ell + s_{\mathtt{r}}$ is the total number of vertices in the graph. We denote an edge of $\mathcal{E}$ as $\{Y_i, Z_j\}$, and its label as $\mathcal{L}(\{Y_i, Z_j\}) = \lambda_{ij}$ (thus, $\lambda_{ij} = \lambda_{ji}$). For a path $\mathcal{P}$ in the graph G, we define the label of the path as $\mathcal{L}(\mathcal{P}) := \sum_{e \in \mathcal{P}} \mathcal{L}(e)$. Similarly, for a cycle $\mathcal{C}$ in the graph G, we define the label of the cycle as $\mathcal{L}(\mathcal{C}) := \sum_{e \in \mathcal{C}} \mathcal{L}(e)$. We say the graph G is **good** if it satisfies the following conditions:

1. G is acylic.
2. Maximum path length of G is two.
3. $\mathcal{L}(\mathcal{P}) \neq \mathbf{0}$, for all paths in G.

For such a good graph G, we associate a system of bivariate affine equations as follows:

$$\mathcal{E}_{\mathsf{G}}^{\mathtt{bi}} := \{Y_i \oplus Z_j = \lambda_{ij} \; \forall \; \{Y_i, Z_j\} \in \mathcal{E}\}.$$

In the above system of bivariate affine equations, the variables are the vertices of the associated graph G. We say that two variables are involved in an equation if the corresponding vertices are connected by an edge in the graph. The constants of the equation are the label of the corresponding edge. Therefore, for $\mathcal{E}_{\mathsf{G}}^{\mathtt{bi}}$, the variables are Y_i's and Z_i's. For a good graph G, two vertices are said to be adjacent to each other if and only if they are connected by an edge in $\mathcal{E}$. This induces a partition on $\mathcal{V}_1 \sqcup \mathcal{V}_2$, and each connected component is called a component. The size of a component refers to the number of elements (i.e., the number of vertices) in the partition, and the set of components is denoted by $\mathsf{comp}(\mathsf{G}) = (\mathsf{C}_1 \sqcup \cdots \sqcup \mathsf{C}_\alpha)$ where each component is of size at least 2.

Definition 1. Let $\mathcal{E}_{\mathsf{G}}$ be a system of equations corresponding to a good acyclic edge-labeled bipartite graph G (as defined above). An injective function $\Phi : \mathcal{V}_1 \sqcup \mathcal{V}_2 \to \{0,1\}^n$, is said to be an *injective solution* to $\mathcal{E}_{\mathsf{G}}$, if for all $\{Y_i, Z_j\} \in \mathcal{E}_{\mathsf{G}}$, it holds that $\Phi(Y_i) \oplus \Phi(Z_j) = \lambda_{ij}$.

In the following, we state a variant of mirror theory, which asserts that if G is a good, acyclic, edge-labelled bipartite graph that can be decomposed into finitely many components of size at least 2, then the number of solutions to $\mathcal{E}_{\mathsf{G}}$ chosen outside of the set $\mathcal{S}_1 \times \mathcal{S}_2 \subseteq \{0,1\}^n \times \{0,1\}^n$, where $\mathcal{S}_1, \mathcal{S}_2$ are two non-empty finite arbitrary subsets of $\{0,1\}^n$, is very close to the average number of solutions until the number of edges in $\mathcal{E}_{\mathsf{G}}$ is roughly $2^{2n/3}$. In the traditional mirror theory, solutions to $\mathcal{E}_{\mathsf{G}}$ is chosen from the set $\{0,1\}^n \times \{0,1\}^n$. However, in the current setup, we choose the solution from a non-empty subset of $\{0,1\}^n \times \{0,1\}^n$. We refer to this variant of mirror theory as *mirror theory over restricted set*. It is worth mentioning that Dutta and Nandi [DN20] have already established a similar security bound for the mirror theory over a restricted set when the graph G is an acyclic good *general* graph. In other words, the authors of [DN20] have shown that the number of solutions to $\mathcal{E}_{\mathsf{G}}$, chosen from a non-empty subset of $\{0,1\}^n$, is close to the average number of solutions until the number of edges in $\mathcal{E}_{\mathsf{G}}$ is roughly $2^{2n/3}$. The following lemma gives a similar lower bound for the mirror theory over a restricted set when the graph G is a good, acyclic, edge-labeled bipartite graph, proof of which can be found in Theorem 1 of [Che22]. For completeness, we provide an independent proof of this lemma in Appendix 7.

Lemma 3. *Let $\mathsf{G} = (\mathcal{V}_1 \sqcup \mathcal{V}_2, \mathcal{E}, \mathcal{L})$ be a good acyclic edge-labelled bipartite graph with s_ℓ many vertices in $\mathcal{V}_1$ and $s_{\mathtt{r}}$ many vertices in $\mathcal{V}_2$, such that $|\mathcal{E}| = q$ and $s = s_\ell + s_{\mathtt{r}}$, the total number of vertices of the graph G. Let the graph G have α components such that the i-th component has c_i vertices from $\mathcal{V}_1$ and d_i vertices from $\mathcal{V}_2$ such that $s_\ell = c_1 + \cdots + c_\alpha$ and $s_{\mathtt{r}} = d_1 + \cdots + d_\alpha$. Let $\rho_i[1] = (c_1 + c_2 + \cdots + c_i)$ and $\rho_i[2] = (d_1 + d_2 + \cdots + d_i)$. Then the total number of injective solutions to $\mathcal{E}_{\mathsf{G}}$ chosen from outside of the set $\mathcal{S}_1 \times \mathcal{S}_2 \subseteq \{0,1\}^n \times \{0,1\}^n$, is at least*

$$\frac{\mathbf{P}(2^n - \Delta_1, s_\ell)\mathbf{P}(2^n - \Delta_2, s_{\mathtt{r}})}{2^{nq}} \cdot \left(1 - \sum_{i=1}^{\alpha} \left(\frac{10\binom{c_i + d_i}{2}\left(\Delta_1 + \Delta_2 + \rho_{i-1}[1] + \rho_{i-1}[2]\right)^2}{2^{2n}} \right)\right),$$

provided $(\Delta_1 + \rho_\alpha[1])c_{\max} \leq 2^{n-3}$ and $(\Delta_2 + \rho_\alpha[2])d_{\max} \leq 2^{n-3}$, where $c_{\max} = \max\{c_1, c_2, \cdots, c_\alpha\}$, $d_{\max} = \max\{d_1, d_2, \ldots, d_\alpha\}$, Δ_1 denotes the size of the set $\mathcal{S}_1$ and Δ_2 denotes the size of the set $\mathcal{S}_2$.

3 (Multi-User) Security Results of LightMAC and LightMAC_Plus

In this section, we first state the existing security results of LightMAC and LightMAC_Plus. In FSE'16, Luykx et al. [LPTY16] have shown that LightMAC is secured against all information-theoretic distinguishers in the single user setting under the pseudorandom permutation assumption of the underlying block cipher of the construction that makes roughly up to $2^{n/2}$ queries such that the maximum length of the message is $2^s(n-s)$ bits, where n is the block size of the underlying block cipher and s is the size of the block counter. The following result establishes an upper bound on the PRF advantage of LightMAC construction against all information-theoretic adversaries in the single-user setting.

Theorem 2 (**SU-Security of LightMAC**). *Let $\mathcal{K}$ be a finite and non-empty set. Let $\mathsf{E} : \mathcal{K} \times \{0,1\}^n \to \{0,1\}^n$ be a block cipher. Then, the PRF advantage for any (q,t) adversary against LightMAC[E] is given by,*

$$\begin{aligned}\mathbf{Adv}^{\mathrm{PRF}}_{\mathsf{LightMAC[E]}}(q,t) &\leq \left(1+\frac{1}{2^{n/2}-1}+\frac{1}{2(2^{n/2}-1)^2}\right)\frac{q^2}{2^n}+\mathbf{Adv}^{\mathrm{PRP}}_{\mathsf{E}}(q(2^s-1),t_1)\\ &+ \mathbf{Adv}^{\mathrm{PRP}}_{\mathsf{E}}(q,t_2),\end{aligned}$$

provided the maximum number of message blocks $\ell \leq 2^{n/2}$ and $t_1 = t + O(q(2^s-1))$, and $t_2 = t + O(q)$.

In ASIACRYPT'17, Naito [Nai17] has shown that LightMAC_Plus is secure against all information-theoretic distinguishers in the single user setting under the pseudorandom permutation assumption of the underlying block cipher of the construction that makes roughly up to $2^{2n/3}$ queries such that the maximum length of the message is $\min\{2^s(n-s),(n-s)2^{n-1}\}$ bits, where n is the block size of the underlying block cipher and s is the size of the block counter. The following result establishes an upper bound on the PRF advantage of LightMAC_Plus against all information-theoretic adversaries in the single-user setting.

Theorem 3 (**SU-Security of LightMAC_Plus**). *Let $\mathcal{K}$ be a finite and non-empty set. Let $\mathsf{E} : \mathcal{K} \times \{0,1\}^n \to \{0,1\}^n$ be a block cipher. Then, the PRF advantage for any (q,t) adversary against LightMAC_Plus[E] is given by,*

$$\mathbf{Adv}^{\mathrm{PRF}}_{\mathsf{LightMAC\_Plus[E]}}(q,t) \leq \frac{2q^2}{2^{2n}}+\frac{4q^3}{2^{2n}}+\mathbf{Adv}^{\mathrm{PRP}}_{\mathsf{E}}(q(2^s-1),t_1)+2\mathbf{Adv}^{\mathrm{PRP}}_{\mathsf{E}}(q,t_2),$$

provided the maximum number of message blocks $\ell \leq 2^{n/2}$ and $t_1 = t + O(q(2^s-1))$, and $t_2 = t + O(q)$.

The bound has been later improved from $2n/3$ bits to $3n/4$ bits in [KLL20].

3.1 Multi User Security Result of LightMAC

In this section, we state the security result of LightMAC in the multi-user setting. In particular, we state and prove that LightMAC is secure against all information-theoretic distinguishers in the multi-user setting under the assumption that the underlying block cipher is an ideal cipher. The following result establishes an upper bound on the multi-user PRF advantage of LightMAC against all information-theoretic adversaries.

Theorem 4 (**MU-Security of LightMAC**)**.** *Let* $\mathsf{E} \leftarrow\!\!\$\ \mathsf{BC}(\{0,1\}^k, \{0,1\}^n)$ *be an ideal block cipher. Then, for any computationally unbounded distinguisher that makes a total of* $q \leq 2^n$ *construction queries across all* u *users, such that each user makes at most* $q_{\max}$ *queries, with* ℓ *being the maximum number of message blocks queried, and a total of* $p \leq 2^k$ *ideal-cipher queries to* E*, can distinguish* LightMAC *from an* n*-bit uniform random function with prf advantage*

$$\begin{aligned}\mathbf{Adv}^{\text{mu-PRF}}_{\mathsf{LightMAC[E]}}(u, q, q_{\max}, p, \ell) \quad &\leq \quad \frac{2u^2}{2^k} + \frac{2qq_{\max}}{2^n} + \frac{28p}{2^k} + \underbrace{\min\left\{\frac{up}{2^k}, \frac{q\ell k}{2^n}\right\}}_{\mathsf{E.1}} \\ &+ \underbrace{\min\left\{\frac{up}{2^k}, \frac{qq_{\max}\ell k}{2^n}, \frac{p\sqrt{q}}{2^k} + \frac{q^{3/2}k\ell}{2^n}\right\}}_{\mathsf{E.2}}.\end{aligned}$$

3.2 Implication of Theorem 4 in the Context of AES:

We instantiate LightMAC with AES-128-256, AES-128-128, and AES-128-192, and derive the corresponding concrete security bound of the LightMAC construction. As per the recommendation of NIST, we consider that the offline data complexity (p) is at least 2^{112}.

1. Instantiating LightMAC with AES-128-256: In this setup, the key size is 256 bits and thus the maximum number of allowed users is 2^{127} due to the term $2u^2/2^k$. Now, if $p \leq 2^{128}$, then we can bound E.1 and E.2 by $up/2^k$ and therefore, we have

$$\mathbf{Adv}^{\text{mu-PRF}}_{\mathsf{LightMAC[E]}}(u, q, q_{\max}, p, \ell) \quad \leq \quad \frac{2u^2}{2^k} + \frac{2qq_{\max}}{2^n} + \frac{28p}{2^k} + \frac{2up}{2^k}.$$

This ensures that we obtain an ℓ-free bound of the construction with major terms $O(up/2^{256} + qq_{\max}/2^{128})$. Note that $q \geq u$ and thus $q \geq 2^{127}$ if we consider $u = 2^{127}$. On the other hand, we have $qq_{\max} \leq 2^{127}$ due to the term $2qq_{\max}/2^n$. This renders the fact that $q_{\max} = 1$ as $n = 128$ due to the term $2qq_{\max}/2^n$. On the other extreme, if we wish the offline data complexity to be more than 2^{128}, then we cannot bound E.1 or E.2 by the term $up/2^k$. In that case, we bound E.1 by $q\ell k/2^n$ and E.2 by $qq_{\max}\ell k/2^n$ and thus, we have

$$\mathbf{Adv}^{\text{mu-PRF}}_{\mathsf{LightMAC[E]}}(u, q, q_{\max}, p, \ell) \quad \leq \quad \frac{2u^2}{2^k} + \frac{2qq_{\max}}{2^n} + \frac{28p}{2^k} + \frac{2qq_{\max}\ell k}{2^n}.$$

Since $q \geq u$ and $u = 2^{127}$, we have $q \geq 2^{127}$, but then $q_{\max}\ell k \leq 1$ which is impossible. Thus, we rule out the choice of having $u \leq 2^{127}$ and $p \geq 2^{128}$. Therefore, we deal with the practical choices of the number of users.

If we consider $u = 2^{64}$, which we believe to be the practical choice of the number of users, the construction achieves security up to a total of 2^{96} construction queries and 2^{192} primitive queries when we restrict the maximum number of allowable queries to 2^{32} per user. If we wish to increase the number of queries per user (say to $2^{40}/2^{48}/2^{56}/2^{64}$), only the total number of construction queries would decrease accordingly ($2^{88}/2^{80}/2^{72}/2^{64}$, respectively) keeping the number of users and the total primitive queries) same. Note that the total number query complexity in the first three cases effectively crosses the usual birthday bound security of the LightMAC construction. Note that, if $u = 1$ (which corresponds to the single user security bound), then the construction achieves security up to a total of 2^{64} construction queries and 2^{251} primitive queries (due to the term $28p/2^k$) when we restrict the maximum number of allowable queries to 2^{64} per user. It is to be noted that this bound matches with the known security bound of the LightMAC construction in the single-user setting.

2. Instantiating LightMAC with AES-128-128: In this setup, the key size is 128 bits and thus the maximum number of allowed users is 2^{63} due to the term $2u^2/2^k$. However, if $p \geq 2^{112}$, then we cannot bound E.1 or E.2 by $up/2^k$. Thus, we bound E.1 by $q\ell k/2^n$ and E.2 by $qq_{\max}\ell k/2^n$. Thus, we have

$$\mathbf{Adv}^{\text{mu-PRF}}_{\text{LightMAC[E]}}(u, q, q_{\max}, p, \ell) \leq \frac{2u^2}{2^k} + \frac{2qq_{\max}}{2^n} + \frac{28p}{2^k} + \frac{2qq_{\max}\ell k}{2^n}.$$

Since $q \geq u$ and $u = 2^{64}$, therefore, $q \geq 2^{64}$. By putting up concrete values, we obtain that the construction is secure up to 2^{64} construction queries and 2^{123} primitive queries if we restrict the maximum number of allowable queries to 2^{32} per user and the maximum number of message blocks being 2^{24}. Similarly, if we further limit the maximum number of queries per user to 2^{24}, then the construction achieves security up to 2^{72} construction queries and 2^{123} primitive queries with the maximum number of message blocks being 2^{24}. However, if we want to allow processing 2^{32} blocks, keeping the offline primitive queries and number of users the same, then we lose on the total number of construction queries (up to 2^{65}) and the maximum number of queries per user (up to 2^{24}).

On the other hand, if the maximum number of users is at most 2^{15}, which we believe to be a practical choice, and the offline data complexity is at most 2^{112}, then E.1 and E.2 can be bounded by $up/2^k$. Thus, we have

$$\mathbf{Adv}^{\text{mu-PRF}}_{\text{LightMAC[E]}}(u, q, q_{\max}, p, \ell) \leq \frac{2u^2}{2^k} + \frac{2qq_{\max}}{2^n} + \frac{28p}{2^k} + \frac{2up}{2^k}.$$

This ensures that we obtain ℓ-free bound of the construction with major terms $O(up/2^{128} + qq_{\max}/2^{128})$. By putting up concrete values, we obtain that the construction is secure up to 2^{72} queries and 2^{112} primitive queries if we restrict the maximum number of queries to 2^{56} per user and the total number of users is 2^{15}. Moreover, if the offline data complexity is more than 2^{112} while the number of users is 2^{15}, then we cannot bound E.1 and E.2 by $up/2^k$. In that case, E.1 would be bounded by $q\ell k/2^n$ and E.2 would be bounded by $qq_{\max}\ell k/2^n$, which boils down to the earlier case with $u = 2^{64}$.

3. Instantiating LightMAC with AES-128-192: In this setup, the key size of the block cipher is 192 bits. Therefore, the maximum number of allowed users is at most 2^{96} due to the term $2u^2/2^k$. Since, the offline data complexity is at least 2^{112}, we cannot bound E.1 or E.2 by $up/2^k$. Therefore, we bound E.1 by $q\ell k/2^n$ and E.2 by $qq_{\max}\ell k/2^n$. Therefore, we have

$$\mathbf{Adv}^{\text{mu-PRF}}_{\text{LightMAC[E]}}(u, q, q_{\max}, p, \ell) \leq \frac{2u^2}{2^k} + \frac{2qq_{\max}}{2^n} + \frac{28p}{2^k} + \frac{2qq_{\max}\ell k}{2^n}.$$

By putting up concrete values, we obtain the construction is secure up to 2^{64} construction queries and 2^{123} primitive queries if we restrict the maximum number of allowable queries to 2^{32} per user with the maximum number of message blocks being 2^{24} and the total number of users is 2^{96}. Similarly, if we further limit the maximum number of queries per user to 2^{24}, then the construction achieves security up to 2^{72} construction queries and 2^{123} primitive queries with the maximum number of message blocks being 2^{24}. However, if we want to allow processing 2^{32} blocks, keeping the offline primitive queries and number of users the same, then we lose on the total number of construction queries (up to 2^{65}) and the maximum number of queries per user (up to 2^{24}). On the other hand, if we consider the number of users

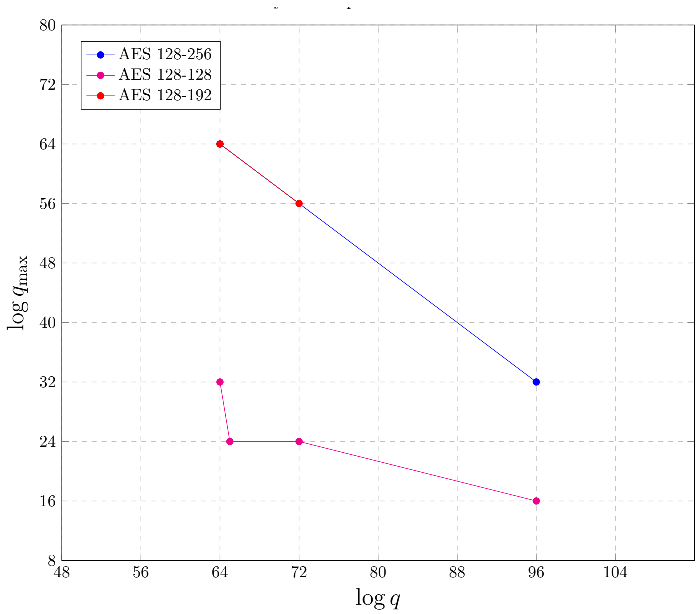

Figure 3: Graphical Representation of the relation between q and $q_{\max}$ for 2^{16} and 2^{32} users with an upper limit on the primitive queries being 2^{96} for the LightMAC construction, which is independent on the bound of the maximum number of message blocks queried.

to be at most 2^{64}, then we can bound E.1 by $q\ell k/2^n$ and E.2 by $p\sqrt{q}/2^k + q^{3/2}k\ell/2^n$. Hence, we have

$$\mathbf{Adv}^{\text{mu-PRF}}_{\mathsf{LightMAC[E]}}(u, q, q_{\max}, p, \ell) \leq \frac{2u^2}{2^k} + \frac{2qq_{\max}}{2^n} + \frac{28p}{2^k} + \frac{p\sqrt{q}}{2^k} + \frac{2q^{3/2}k\ell}{2^n}.$$

By putting up concrete values, we obtain the construction is secure up to 2^{64} construction queries and 2^{160} primitive queries if we restrict the maximum number of allowable queries to 2^{64} per user with the maximum number of message blocks being 2^{24} and the total number of users is 2^{64}. Similarly, if we further limit the maximum number of queries per user to 2^{56}, then the construction achieves security up to 2^{72} construction queries and 2^{156} primitive queries with the maximum number of message blocks being 2^{12}. If we reduce the number of users to 2^{58}, then the construction achieves security up to 2^{58} construction queries with 2^{163} primitive queries, with a maximum number of queries per user of 2^{58} while the maximum number of message blocks is 2^{32}.

We depict the security of the LightMAC construction in terms of the number of users, total number of construction queries and offline queries, maximum number of queries per user, and the maximum number of message blocks in various use cases in Table 1. We also demonstrate the variation of q vs $q_{\max}$ for AES-128-256, AES-128-192, and AES-128-128 when the number of users is 2^{16} and 2^{32} in Fig. 3 and Fig. 4.

3.3 Multi User Security Result of LightMAC_Plus

Now, we state the security result of the LightMAC_Plus construction in the multi-user setting. In particular, we state and prove that LightMAC_Plus is secure against all information-theoretic distinguishers in the multi-user setting under the assumption that the underlying block cipher is an ideal cipher that makes roughly up to $2^{2n/3}$ many message

Table 1: Security of LightMAC construction in terms of the number of users u, number of construction queries q, maximum number of queries per user $q_{\max}$, maximum number of message blocks ℓ and the total number of offline primitive queries p. $\star$ in the column labeled with ℓ denotes that the corresponding security bound is independent of the maximum message length.

Block Cipher	u	q	$q_{\max}$	ℓ	p
AES-128-256	2^{127}	2^{127}	1	$\star$	2^{128}
	2^{64}	2^{96}	2^{32}	$\star$	2^{192}
	2^{64}	2^{64}	2^{64}	$\star$	2^{192}
	1	2^{64}	2^{64}	$\star$	2^{251}
AES-128-128	2^{63}	2^{64}	2^{32}	2^{24}	2^{123}
	2^{63}	2^{72}	2^{24}	2^{24}	2^{123}
	2^{63}	2^{65}	2^{24}	2^{32}	2^{123}
	2^{15}	2^{72}	2^{56}	$\star$	2^{112}
AES-128-192	2^{96}	2^{64}	2^{32}	2^{24}	2^{123}
	2^{96}	2^{72}	2^{24}	2^{24}	2^{123}
	2^{96}	2^{65}	2^{24}	2^{32}	2^{123}
	2^{64}	2^{64}	2^{64}	2^{24}	2^{160}
	2^{64}	2^{72}	2^{56}	2^{12}	2^{156}
	2^{58}	2^{58}	2^{58}	2^{32}	2^{163}

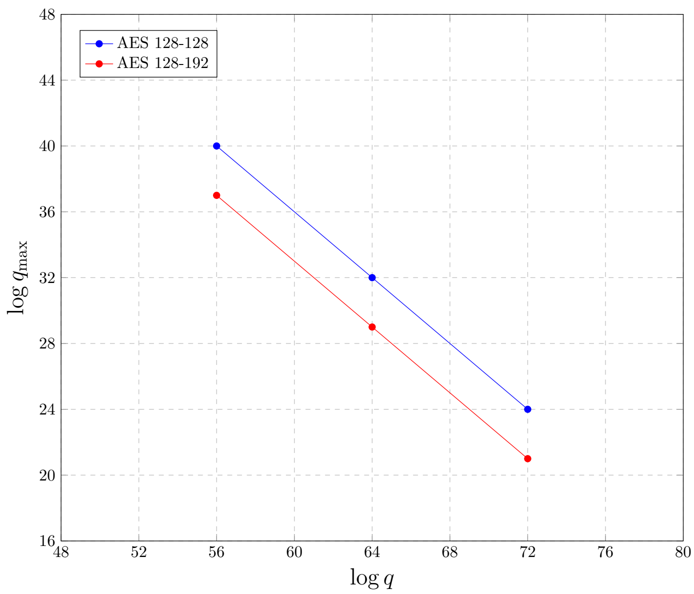

Figure 4: Graphical Representation of the relationship between q and $q_{\max}$ for 2^{16} and 2^{32} users with a lower limit on the primitive queries being 2^{176} and the maximum number of message blocks is 2^{24} for the LightMAC construction.

queries and roughly $2^{2k/3}$ many ideal-cipher queries such that the maximum number of message blocks is at most $2^s(n-s)$ bits, where n is being the block size of the underlying block cipher and s is the size of the block counter. The following result establishes an upper bound on the multi-user PRF advantage of LightMAC_Plus construction against all information-theoretic adversaries.

Theorem 5 (**MU Security of LightMAC_Plus**)**.** *Let $\mathcal{K}$ be a finite and non-empty set. Let $\mathsf{E} \leftarrow\!\$\ \mathsf{BC}(\mathcal{K}, \{0,1\}^n)$ be a ideal block cipher. Then any A computationally unbounded distinguisher, making a total of q construction queries across all u users such that each user makes at most $q_{\max}$ queries with the maximum number of message blocks in a query is ℓ, and a total of p ideal-cipher queries to the ideal block cipher* E*, can distinguish* LightMAC_Plus *from an n-bit uniform random function with prf advantage*

$$\begin{aligned}\mathbf{Adv}^{\text{mu-PRF}}_{\mathsf{LightMAC\_Plus[E]}}(u,q,q_{\max},p,\ell) &\leq \tfrac{2u}{2^k} + \tfrac{u^3}{6.2^{2k}} + \tfrac{2up^2}{2^{2k}} + \tfrac{6q^2\ell}{2^{n+k}} + \tfrac{5qq_{\max}^2}{2^{2n}} + \tfrac{2p\sqrt{q}}{2^k} \\ &+ \tfrac{2q^{3/2}k\ell}{2^n} + \tfrac{2q\sqrt{q_{\max}}}{2^n} + \min\left\{\tfrac{2up}{2^k}, \tfrac{2q\ell k}{2^n}\right\} + \tfrac{52p}{2^k} \\ &+ \tfrac{40p^2q^2}{2^{3n}} + \tfrac{320(p+q)q^3}{2^{3n}} + \tfrac{160p^2q}{2^{2n}} + \tfrac{1280(p+q)q^2}{2^{2n}},\end{aligned}$$

where $4q < 2^n$, $6p < 2^k$ and $\ell \leq \min\{2^{n-1}, 2^s\}$.

Note that the above security bound of LightMAC_Plus contains a factor of ℓ, whereas the single-user security of the LightMAC_Plus construction possesses an ℓ-free bound, provided $\ell \leq 2^{n/2}$. We show that if we assume that $k \geq 4n/3$, then we can prove an ℓ-free multi-user security bound of the LightMAC_Plus construction, provided $\ell \leq 2^{n/3}/k$. Formally, we have the following:

Theorem 6 (**ℓ-free MU Security of LightMAC_Plus**)**.** *Let $\mathcal{K}$ be a finite and non-empty set. Let $\mathsf{E} \leftarrow\!\$\ \mathsf{BC}(\mathcal{K}, \{0,1\}^n)$ be a ideal block cipher. Then any computationally unbounded distinguisher, making a total of q construction queries across all u users such that each user makes at most $q_{\max}$ queries and a total of p ideal-cipher queries to the ideal block cipher* E*, can distinguish* LightMAC_Plus *from an n-bit uniform random function with prf advantage*

$$\begin{aligned}\mathbf{Adv}^{\text{mu-PRF}}_{\mathsf{LightMAC\_Plus[E]}}(u,q,q_{\max},p) &\leq \tfrac{2u}{2^k} + \tfrac{u^3}{6.2^{2k}} + \tfrac{2up^2}{2^{2k}} + \tfrac{5q^2}{n2^{2n}} + \tfrac{5qq_{\max}^2}{2^{2n}} + \tfrac{2p\sqrt{q}}{2^k} \\ &+ \tfrac{2q^{3/2}}{2^n} + \tfrac{2q\sqrt{q_{\max}}}{2^n} + \min\left\{\tfrac{2up}{2^k}, \tfrac{2q}{2^{2n/3}}\right\} + \tfrac{13p}{2^{3k/4}} \\ &+ \tfrac{40p^2q^2}{2^{3n}} + \tfrac{320(p+q)q^3}{2^{3n}} + \tfrac{160p^2q}{2^{2n}} + \tfrac{1280(p+q)q^2}{2^{2n}},\end{aligned}$$

where $q \leq 2^{n-2}$, $k \geq 4n/3$ and ℓ is the maximum number of message blocks queried such that $\ell \leq \min\{2^{n/3}/k, 2^s\}$.

3.4 Implication of Theorem 5 and 6 in the Context of AES

If we instantiate LightMAC_Plus with AES-128-128, we can only apply Theorem 5. In this case, the construction provides security up to 2^{64} construction queries with a maximum allowable message length of 2^{24} blocks. The maximum primitive queries allowed in this case is 2^{92}. If the maximum message length is 2^{12}, the construction provides beyond the birthday bound security up to 2^{72} construction queries and 2^{88} primitive queries. If we instantiate LightMAC_Plus with AES-128-192 or AES-128-256, we can apply Theorem 6 which ensures that the construction provides security up to 2^{84} construction queries and 2^{82} primitive queries. This bound allows the maximum message length to be of size 2^{34} blocks. In addition, the bound holds for any number of users and any maximum number of queries per user, satisfying that the total number of construction queries is less than 2^{84}.

4 A Generic Overview of the Proof Approach

In this section, we briefly describe the approach of the multi-user security proof of the LightMAC and LightMAC_Plus construction as follows:

4.1 Proof Approach of LightMAC

We prove the security of the construction using the H-Coefficient technique [Pat08], where we release the block cipher keys of all the users and all the intermediate output values V in both worlds, along with the usual query-response pairs after the interaction is over. This allows the challenger in both worlds to compute Σ values for defining bad transcripts in terms of Σ and the released keys. Note that LightMAC uses two keys K_1 and K_2, where K_1 is used to compute the Σ values, which we customarily refer to as *hash key*, and K_2 is used to compute the tags, which are customarily referred to as *prf key*.

1. If the prf-key and a tag value (resp. Σ value) of an user collides with an ideal cipher key and the output value (resp. input value) of the corresponding ideal-cipher query, then in the real world, the corresponding Σ (resp. tag) value of that user will match with the input (resp. output) of the corresponding ideal-cipher query with probability 1, but that event holds in the ideal world with low probability, which in turn allows a distinguisher to distinguish between the real and the ideal world. Hence, we consider this event to be bad (see Bad1 and Bad2 of Sect. 5.2). A pictorial description of the events is shown in the top and bottom figures of Fig. 5.

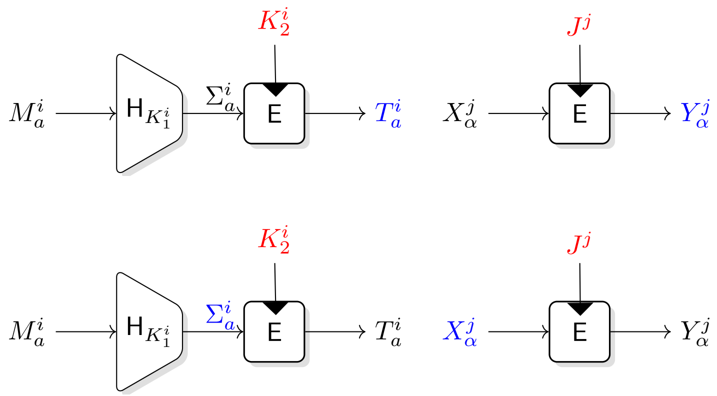

Figure 5: Illustration of events Bad1 and Bad2 where K_2^i matches with an ideal cipher key J^j in both cases. The top (resp. bottom) figure depicts a collision between the response (resp. input) to the a-th query and the output Y_α^j (resp. input X_α^j) of the ideal-cipher query.

2. We disregard the collision between two Σ values for any user (see Bad3 of Sect. 5.2). A pictorial description of the event is shown in Fig. 6.

3. No two users should share the same keys. Each user must be given a fresh hash key and PRF key. If two users end up with the same hash key, the same PRF key, or if one user's hash key matches another user's PRF key, we treat these as bad events (see BadK1, BadK2 in Sect . 5.2). We want to mention that the freshness of keys will help us to lower-bound the ratio of real to ideal interpolation probability for good transcripts.

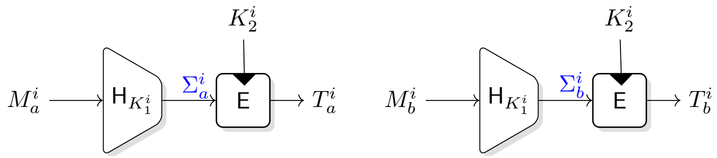

Figure 6: Description of the event Bad3: Σ values of two queries for a user collide with each other.

4. To simplify the analysis of the bad event Bad1, we require that every tag (response) generated for any user be unique. If two tags for the same user i are found to be equal, we set the bad event BadCollT to true.

5. To support the analysis of Bad2 and Bad3, we introduce an auxiliary bad event where a user's hash key collides with an ideal cipher key. If such a collision occurs, we treat it as a bad event (see $\mathsf{Bad}_{\mathsf{Aux}_i}$ in Sect. 5.2).

If the bad events do not hold, then we count the total number of block cipher calls to lower bound the real interpolation probability. Computing the ideal interpolation probability is easy, and thus we show that the ratio of the former to the latter is at least 1. However, the non-triviality of the proof lies in upper-bounding the probability of some crucial bad events.

Since we are proving the security of the constructions in the ideal-cipher model, it is plausible to assume that the adversary is well-informed about all the V values (during the block cipher evaluations) that define the event $\Sigma_a^i = \Sigma_b^i$. In that case, we are no longer left with any randomness that will bound the bad event Bad3. However, the good news is that should the above situation hold, it must be the case that the i-th user key K_1^i collides with a chosen ideal-cipher key J^j, which contributes to 2^{-k} probability to the event. Now, if one varies over all possible choices of indices (i, a, b, j), the probability becomes $u^2p/2^k$, making our bound worse. The trick that we apply here is that we do not allow the choices of j to vary up to p. We restrict this choice to some parameter μ, which is of the order $2^{k-n}k\ell$, where ℓ denotes the maximum number of message blocks. For this choice of μ, we restore our target bound, at the cost of introducing an extra event that dictates the number of j that exceeds μ holds with a low probability. Details of the analysis can be found in Sect. 5.2.

4.2 Proof Approach of LightMAC_Plus

We prove the security of the LightMAC_Plus construction using the well known Expectation Method [HT16], where we release the block cipher keys of all the users and all the intermediate output values V in both worlds along with the usual query-response pairs after the interaction is over. This allows the challenger in both worlds to compute (Σ, Θ) values for defining bad transcripts in terms of it and the released keys. Note that LightMAC_Plus uses three keys K_1, K_2 and K_3, where K_1 is used to compute the (Σ, Θ) values, which we customarily refer to as *hash key* and K_2, K_3 is used to compute the tags. We refer to K_2 as *σ-prf key* and K_3 as *θ-prf key*.

We first impose some bad conditions on user keys, which mainly says that

- a pair of keys should not collide for two different users (see BadK1-BadK3 of Sect. 6.1).

- Moreover, if the prf key collides for two different users, then none of their corresponding hash keys should collide with any ideal-cipher keys (see BadK4 of Sect. 6.1).

- If both the prf keys of a user collide with ideal-cipher keys (see BadK5 of Sect. 6.1), then we set it bad.

- Moreover, both the prf keys of a user should not collide with each other (see BadK6 of Sect. 6.1).

- Finally, if the hash key of a user i collides with σ-prf key (resp. θ-prf key) of an another user i', then none of the message blocks preprended with the appropriate block counter of user i should not collide with Σ (resp. Θ) value of user i' (see BadK7, BadK8 of Sect. 6.1).

Following the definition of bad events on user keys, we now define bad events based on the input and key collision as follows:

1. If σ-prf key (resp. θ-prf key) for any user collides with an ideal cipher key, then its Σ (resp. Θ) value should not collide with the corresponding ideal-cipher input. Otherwise, in the real world, one can determine the block cipher output of Σ (resp. Θ), which does not hold in the ideal world (see Bad1, Bad2 of Sect. 6.1).

2. If the hash key for any user collides with an ideal-cipher key, then its corresponding Σ (or Θ) value should not collide with other Σ (or Θ) values (see Bad3, Bad4 of Sect. 6.1).

3. If two Σ (resp. Θ) value of a user collides with each other, then their corresponding tag value should be distinct (see Bad5, Bad6 of Sect. 6.1).

4. If a pair of σ (resp. θ)-prf keys collide with each other, then their corresponding Σ value or Θ value should not collide with each other (see Bad7-Bad10 of Sect. 6.1).

5. If two Σ (resp. Θ) values of a user collide with each other, then the corresponding Θ (resp. Σ) value should be distinct (see Bad11 of Sect. 6.1), and should not collide with any other Θ value of the same user (see Bad12 of Sect. 6.1).

6. Finally, the number of colliding Σ values or Θ values for an user i should be at most $q_i^{1/2}$, where q_i denotes the number of queries of the i-th user (see Bad13, Bad14).

If the bad events do not happen, then we lower bound the ratio of the real to ideal interpolation probability. Bounding the ideal interpolation probability is straightforward. To lower bound the real interpolation probability, we do the following: we divide the set of users into two classes: (i) a set of users whose exactly one of the prf keys collides with an ideal-cipher key and (ii) set of users whose none of the prf keys collide with any ideal-cipher key. For the first class of users, we further divide them into a finite number of partitions based on the relation that two users belong to the same class if their corresponding prf keys collide with the same ideal-cipher key. Then, we count the number of solutions to the system of equations by using the result of mirror theory over the restricted set:

$$\mathsf{E}_{K_2^i}(\Sigma_a^i) \oplus \mathsf{E}_{K_3^i}(\Theta_a^i) = T_a^i.$$

Similarly, for the second class of users, we further divide them into a finite number of partitions based on the relation that two users belong to the same equivalent classes, if their corresponding prf keys collide with each other and then count the number of solutions to the system of equations by applying the result of mirror theory:

$$\mathsf{E}_{K_2^i}(\Sigma_a^i) \oplus \mathsf{E}_{K_3^i}(\Theta_a^i) = T_a^i.$$

However, the non-triviality of the analysis lies in bounding the event $\Sigma_a^i = \Sigma_b^i, \Theta_a^i = \Theta_b^i$ for some user i when its hash key collides with an ideal cipher key. Such an event holds with probability 2^{-k}. Now, if one varies over all possible choices of (i, a, b, j), the probability becomes $q^2p/2^k$, making our bound worse. Thus, we apply the previous trick, i.e., we restrict the choices of j to vary up to some parameter μ and get the bound $\sum_{i=1}^{u} q_i^2\mu/2^k$. By choosing $\mu = 2^{k-n}k\ell$, we obtain the bound $2^{-n}(q_1^2 + \ldots + q_u^2)$. Note that a crude bound on the sum $(q_1^2 + \ldots + q_u^2)$ is q^2 makes us lose our target security bound. Therefore, we carefully bound the sum. We split it into two cases: (a) when $i \leq \sqrt{q}$, then we achieve the desired security bound $p\sqrt{q}/2^k$. On the other hand, (b) when $i \geq \sqrt{q}$, then by applying a combinatorial lemma (Lemma 2), we bound the sum $(q_1^2 + \ldots + q_u^2)$ to be at most $q^{3/2}$ and then by plugging-in the appropriate value of $\mu = 2^{k-n}k\ell$, we obtain the desired security bound. Summarizing above, we would like to mention that to achieve the security bounds of both constructions, our analysis has crucially relied on a combinatorial result stated in Lemma 2.

5 Proof of Theorem 4

We consider a computationally unbounded non-trivial deterministic distinghisher A that interacts with a pair of oracles in either the real world or the ideal world, described as follows: in the real world, A is given access to u independent instances of the $\mathsf{LightMAC}$ construction, i.e., to a tuple of u oracles $(\mathsf{LightMAC}[\mathsf{E}]_{(K_1^i, K_2^i)})_{i \in [u]}$, where each (K_1^i, K_2^i) is independent of (K_1^j, K_2^j) and $\mathsf{E} \leftarrow\!\!\$\ \mathsf{BC}(\mathcal{K}, \{0,1\}^n)$ is an ideal block cipher. Additionally, A has access to the oracle $\mathsf{E}^{\pm}$, underneath the construction $\mathsf{LightMAC}$. In the ideal world, A is given access to (i) a tuple of u independent random functions $(\mathsf{RF}_1, \ldots, \mathsf{RF}_u)$, where each RF_i is the random function from $\{0,1\}^*$ to $\{0,1\}^n$ that can be equivalently described as a procedure that returns an n-bit uniform string on input of any arbitrary message, and (ii) the oracle $\mathsf{E}^{\pm}$, where $\mathsf{E} \leftarrow\!\!\$\ \mathsf{BC}(\mathcal{K}, \{0,1\}^n)$ is an ideal block cipher, sampled independent of the distribution of the sequence of the u independent random functions. In both worlds, the first oracle is called the *construction oracle* and the latter, the *ideal cipher oracle*. Using the ideal cipher oracle, a distinguisher A can evaluate any query x under its chosen key J. A query to the construction oracle is called a *construction query*, and to that of the ideal cipher oracle is called an *ideal cipher query*. Note that A can make either *forward* (i.e., it evaluates E with a chosen key and input), or *inverse* ideal cipher queries (i.e., it evaluates E^{-1} with a chosen key and input).

5.1 Attack Transcript

We summarize the interaction between the distinguisher and the challenger in a transcript $\tau_c = \tau_c^1 \cup \tau_c^2 \cup \ldots \cup \tau_c^u$, where $\tau_c^i = \{(M_1^i, T_1^i), \ldots, (M_{q_i}^i, T_{q_i}^i)\}$ denotes the query-response transcript generated from the i-th instance of the construction. Moreover, we assume that A has chosen g distinct ideal cipher keys $J^1, \ldots, J^g$ such that it makes p_j ideal cipher queries to the block cipher with the chosen key J^j. We summarize the ideal cipher queries in a transcript $\tau_p = \tau_p^1 \cup \tau_p^2 \cup \ldots \cup \tau_p^g$, where $\tau_p^j = \{(u_1^j, v_1^j), \ldots, (u_{p_j}^j, v_{p_j}^j), J^j\}$ denotes the transcript of the ideal cipher queries when the chosen ideal cipher key is J^j. We assume that A makes q_i construction queries for the i-th instance and p_j ideal cipher queries (including forward and inverse queries) with chosen ideal cipher key J^j. We also assume that the total number of construction queries across u instances is q, i.e., $q = (q_1 + \ldots + q_u)$ and the total number of ideal cipher queries is $p = (p_1 + \ldots + p_g)$. Since A is non-trivial, none of the transcripts contains duplicate elements.

We modify the experiment by releasing internal information to A after it has finished its interaction but has not yet output the decision bit. In the real world, we reveal all the keys (K_1^i, K_2^i) for all u users and also the $(V_a^i[\alpha])_{i\in[q],a\in[q_i],\alpha\in[\ell_a^i-1]}$ tuple, where ℓ_a^i is the number of message blocks corresponding the a-th query of the i-th user and $V_a^i[\alpha] = \mathsf{E}_{K_1^i}(\langle\alpha\rangle_s \| M_a^i[\alpha])$. In the ideal world, the challenger sample the keys (K_1^i, K_2^i) uniformly at random from their respective key spaces for all u users and computes the tuple $(V_a^i[\alpha])_{i\in[q],a\in[q_i],\alpha\in[\ell_a^i-1]}$ tuple as follows:

$$V_a^i[\alpha] = \mathsf{E}_{K_1^i}(\langle\alpha\rangle_s \| M_a^i[\alpha])$$

and reveal them to the distinguisher. Therefore, each transcript τ_i^c, where $i \in [u]$, is now modified to include the corresponding pair of keys and $V_a^i[\alpha]$ values for the i-th instance of the construction. Thus,

$$\tilde{\tau}_c^i = \{(M_1^i, T_1^i, \widetilde{V}_1^i), \ldots, (M_{q_i}^i, T_{q_i}^i, \widetilde{V}_{q_i}^i), K_1^i, K_2^i\},$$

where $\widetilde{V}_a^i := (V_a^i[\alpha])_{\alpha\in[\ell_a^i-1]}$. In all the following, the complete construction query transcript is

$$\tau_c = \bigcup_{i=1}^{u} \tilde{\tau}_c^i$$

and the overall transcript is $\tau = \tau_c \cup \tau_p$. The modified experiment only makes the distinguisher more powerful and hence the distinguishing advantage of A in this experiment is no less than its distinguishing advantage in the former. To prove the security of the construction using the H-coefficient technique, we need to identify the set of bad transcripts and compute an upper bound for their probability in the ideal world. Then, we find a lower bound for the ratio of the real to ideal interpolation probability for a good transcript. Therefore, it only remains to bound the probability of bad transcripts in the ideal world and provide a lower bound for the ratio of the real to ideal interpolation probability for a good transcript. It follows that for each $i \in [u]$, $\mathsf{LightMAC[E]}_{(K_1^i, K_2^i)} \mapsto \tilde{\tau}_c^i$ denotes the following:

(a) $\Sigma_a^i = \bigoplus_{\alpha=1}^{\ell_a^i-1} V_a^i[\alpha] \oplus (M_a^i[\ell_a^i] \| 10^\star)$, $a \in [q_i]$

(b) $\mathsf{E}_{K_2^i}(\Sigma_a^i) = T_a^i$, $a \in [q_i]$.

5.2 Definition and Bounding Probability of Bad Transcripts

In this section, we define and bound the probability of bad transcripts in the ideal world. Note that, following (a), one can derive $\widetilde{\Sigma}^i := (\Sigma_1^i, \Sigma_2^i, \ldots, \Sigma_{q_i}^i)$ tuple from the transcript $\tau = (\tau_c, \tau_p)$ for each user i. We denote the input to the α-th ideal-cipher query corresponding to the ideal-cipher key J^j as X_α^j and the corresponding response as Y_α^j. We say that an attainable transcript $\tau = (\tau_c, \tau_p)$ is a **bad** transcript if any one of the following events hold true

1. $\mathsf{BadK1}$: $\exists i_1 \neq i_2 \in [u]$ such that $K_2^{i_1} = K_2^{i_2}$ or $K_1^{i_1} = K_1^{i_2}$
2. $\mathsf{BadK2}$: $\exists i_1 \neq i_2 \in [u]$ such that $K_1^{i_1} = K_2^{i_2}$
3. $\mathsf{BadCollT}$: $\exists i \in [u]$, $a, b \in [q_i]$ such that $T_a^i = T_b^i$
4. $\mathsf{Bad1}$: $\exists i \in [u], a \in [q_i], j \in [g], \alpha \in [p_j]$ such that $K_2^i = J^j, T_a^i = Y_\alpha^j$.
5. $\mathsf{Bad2}$: $\exists i \in [u], a \in [q_i], j \in [g], \alpha \in [p_j]$ such that $K_2^i = J^j, \Sigma_a^i = X_\alpha^j$.
6. $\mathsf{Bad3}$: $\exists i \in [u], a \neq b \in [q_i]$ such that $\Sigma_a^i = \Sigma_b^i$.

Recall that $\mathsf{BadT} \subseteq \mathcal{V}$ is the set of all attainable bad transcripts and $\mathsf{GoodT} = \mathcal{V} \setminus \mathsf{BadT}$ is the set of all attainable good transcripts. Before we proceed to bound the above events in the ideal world, we state the following lemma from [LPTY16, Proposition 1] that upper bounds the collision probability between two Σ values for two distinct queries. We emphasize that the following result will be frequently used in upper-bounding the probability of the above bad events.

Lemma 4 (**Collision of** Σ)**.** *Let M_a and M_b be two distinct messages such that the maximum number of message blocks is ℓ. Then, for an arbitrary constant $c \in \{0,1\}^n$, we have*

$$(i)\ \Pr[\Sigma_a = \Sigma_b] \leq \frac{2}{2^n}, (ii)\ \Pr[\Sigma_a = c] \leq \frac{2}{2^n},$$

provided $\ell \leq 2^{n/2}$. [2]

The following lemma provides an upper bound on the probability of obtaining a bad transcript in the ideal world.

Lemma 5 (**Bad Lemma**)**.** *Let $\tau = (\tau_c, \tau_p)$ be any attainable transcript. Let X_{id} and BadT be defined as above. Then*

$$\begin{aligned}\Pr[\mathsf{X}_{\mathrm{id}} \in \mathsf{BadT}] \quad \leq \quad & \frac{2u^2}{2^k} + \frac{2qq_{\max}}{2^n} + \frac{28p}{2^k} + \min\left\{\frac{up}{2^k}, \frac{q\ell k}{2^n}\right\} \\ & + \min\left\{\frac{up}{2^k}, \frac{qq_{\max}\ell k}{2^n}, \frac{p\sqrt{q}}{2^k} + \frac{q^{3/2}k\ell}{2^n}\right\},\end{aligned}$$

where ℓ is the maximum number of message blocks queried.

Proof. We define $\mathsf{BadT} := \mathsf{BadK1} \vee \mathsf{BadK2} \vee \mathsf{BadCollT} \vee (\mathsf{Bad1} \mid \overline{\mathsf{BadCollT}}) \vee \mathsf{Bad2} \vee \mathsf{Bad3}$. We upper bound the probability of individual bad events in the ideal world. Then, by the union bound, we sum up the bounds to obtain the overall bound on the probability of bad transcripts in the ideal world.

II. Bounding $\mathsf{BadK2}$. Recall that the event $\mathsf{BadK1}$ holds if there exists two distinct users i_1 and i_2 such that $K_2^{i_1}$ collides with $K_2^{i_2}$ or $K_1^{i_1}$ collides with $K_1^{i_2}$. In the ideal world, since the block cipher keys of each user are drawn independently and uniformly at random, for a fixed choice of two users i_1 and i_2, the probability that $K_2^{i_1} = K_2^{i_2}$ holds is exactly 2^{-k}. Similarly, for a fixed choice of two users i_1 and i_2, the probability that $K_1^{i_1} = K_1^{i_2}$ holds, is exactly 2^{-k}. Therefore, by varying over all possible choices of users, we have

$$\Pr[\mathsf{BadK1}] \leq \frac{2\binom{u}{2}}{2^k} \leq \frac{u^2}{2^k}. \tag{4}$$

[2]The bound on the maximum message length in terms of the number of blocks is actually $\min\{2^{n-1}, 2^s\}$. However, as we choose $s = n/2$ for achieving rate $1/2$, the bound on ℓ is at most $2^{n/2}$.

III. Bounding $\mathsf{BadK2}$. Consider the event $\mathsf{BadK2}$, defined as the presence of two distinct users, denoted as i_1 and i_2, such that their corresponding block cipher keys $K_1^{i_1}$ and $K_2^{i_2}$ collide. Since in the ideal world, the block cipher keys for each user are sampled independently and uniformly at random, the probability of $K_1^{i_1}$ being equal to $K_2^{i_2}$ for a fixed choice of i_1 and i_2 is precisely 2^{-k}. Consequently, by considering all possible combinations of users, the probability of the event $\mathsf{BadK2}$ occurring is given by the inequality:

$$\Pr[\mathsf{BadK2}] \leq \frac{\binom{u}{2}}{2^k} \leq \frac{u^2}{2^{k+1}}. \tag{5}$$

IV. Bounding $\mathsf{BadCollT}$. Consider the event $\mathsf{BadCollT}$ holds if there exists a user i such that T values of one of its queries collide with a T value of another query. For a fixed choice of the user i, the probability that $T_a^i = T_b^i$ holds is exactly 2^{-n}. Therefore, by varying over all possible choices of users and their queries, we have

$$\Pr[\mathsf{BadCollT}] \leq \sum_{i=1}^{u} \frac{\binom{q_i}{2}}{2^n} \leq \sum_{i=1}^{u} \frac{q_i^2}{2^{n+1}} \leq q_{\max} \sum_{i=1}^{u} \frac{q_i}{2^{n+1}} \leq \frac{q_{\max} q}{2^{n+1}}. \tag{6}$$

V. Bounding $\mathsf{Bad1} \mid \overline{\mathsf{BadCollT}}$. Note that the event $\mathsf{Bad1}$ holds if there exists an user i such that its block cipher key K_2^i collides with some chosen ideal-cipher key J^j, for some $j \in [g]$, and one of its obtained response T_a^i collides with a Y_α^j value of the corresponding ideal-cipher query. Recall that, in the ideal world, the responses of the user's queries are sampled independently and uniformly at random from $\{0,1\}^n$, and the block cipher keys of every user are sampled from $\mathcal{K}$ uniformly and independently of the distribution of the responses. Also, due to $\overline{\mathsf{BadCollT}}$, all the responses to the user's construction queries are distinct. Now, depending on the order of occurrence of the construction queries and the ideal-cipher queries, we have the following cases:

☐ <u>CASE A: CONSTRUCTION QUERY FOLLOWED BY IDEAL-CIPHER QUERY:</u>
For a fixed choice of $j \in [g]$ and a fixed choice of ideal-cipher query $\alpha \in [p_j]$, the probability that $K_2^i = J^j, T_a^i = Y_\alpha^j$ holds, becomes $2^{-(n+k)}$. Therefore, by varying over all possible choices of indices, we have

$$\Pr[\mathsf{Bad1} \mid \overline{\mathsf{BadCollT}}] \leq \sum_{i=1}^{u} \sum_{a=1}^{q_i} \sum_{j=1}^{g} \sum_{\alpha=1}^{p_j} \frac{1}{2^{n+k}} \leq \frac{pq}{2^{n+k}}. \tag{7}$$

☐ <u>CASE B: IDEAL-CIPHER QUERY FOLLOWED BY CONSTRUCTION QUERY:</u>
If the ideal-cipher query is forward, then a similar analysis as that of CASE A follows. On the other hand, if the ideal-cipher query is backward, we can set Y_α^j to the tag value T_a^i. Therefore, we cannot use the randomness of T and the probability that $K_2^i = J^j, T_a^i = Y_\alpha^j$ holds, becomes 2^{-k}. However, due to $\overline{\mathsf{BadCollT}}$, there exists exactly one choice of (i, a) such that $T_a^i = Y_\alpha^j$. Therefore, by varying over all possible choices of indices, we have

$$\Pr[\mathsf{Bad1} \mid \overline{\mathsf{BadCollT}}] \leq \sum_{j=1}^{g} \sum_{\alpha=1}^{p_j} \frac{1}{2^k} \leq \frac{p}{2^k}. \tag{8}$$

By combining Eqn. (7) and Eqn. (8), and by assuming $q \leq 2^n$, we have

$$\Pr[\mathsf{Bad1} \mid \overline{\mathsf{BadCollT}}] \leq \frac{2p}{2^k}. \tag{9}$$

VI. Bounding $\mathsf{Bad2}$. The event $\mathsf{Bad2}$ holds if there exists an user i such that its block cipher key K_2^i collides with some chosen ideal-cipher key J^j, for some $j \in [g]$, and one of its Σ values Σ_a^i collides with a corresponding primitive query input X_α^j. To bound this event, for each $i \in [u]$, we define an auxiliary event as follows:

$$\mathsf{Bad}_{\mathsf{Aux}_i} := \exists j' \in [g] : K_1^i = J^{j'}.$$

Now, we write the probability of the event Bad1 as follows:

$$\begin{aligned}\Pr[\mathsf{Bad2}] \quad &\leq \quad \sum_{i=1}^{u} \Big(\Pr[\underbrace{\exists a \in [q_i], j \in [g], \alpha \in [p_j] : K_2^i = J^j, \Sigma_a^i = X_\alpha^j, \mathsf{Bad}_{\mathsf{Aux}_i}]}_{\mathsf{E.1}} \Big) \\ &+ \quad \sum_{i=1}^{u} \Big(\Pr[\underbrace{\exists a \in [q_i], j \in [g], \alpha \in [p_j] : K_2^i = J^j, \Sigma_a^i = X_\alpha^j, \overline{\mathsf{Bad}_{\mathsf{Aux}_i}}]}_{\mathsf{E.2}} \Big).\end{aligned}$$

A. Bounding Event E.1. The event E.1 implies that the block cipher key K_1^i of the i-th user collides with some chosen ideal-cipher key $J^{j'}$. As a result, all the V values for computing Σ_a^i may be pre-determined, as the distinguisher can make appropriate ideal-cipher queries to determine all the V values. Hence, to upper bound the probability that $\Sigma_a^i = X_\alpha^j$, we cannot use the randomness of V variables, which in turn prohibits us from applying Lemma 4 to upper bound the probability of $\Sigma_a^i = X_\alpha^j$, for a fixed choice of i, a, j, and α. Therefore, we have to use the event that $K_1^i = J^{j'}$, for some $j' \in [g]$. Now, we bound the probability of the event E.1 in the following two cases depending on the size of the underlying block cipher key.

(a) If the key size of the underlying block cipher is sufficiently large, e.g., 256 bits, then we bound the probability of the event in a very crude way. We bound the event $K_2^i = J^j, \Sigma_a^i = X_\alpha^j, K_1^i = J^{j'}$ to at most 2^{-2k} by using the independence of two random variables K_1^i and K_2^i. However, by varying all possible choices of indices, we have roughly $up^2/2^{2k}$ bound. By assuming $p \leq 2^k$, we have

$$\sum_{i=1}^{u} \Pr\left[\exists a \in [q_i], j \in [g], \alpha \in [p_j] : K_2^i = J^j, \Sigma_a^i = X_\alpha^j, \mathsf{Bad}_{\mathsf{Aux}_i}\right] \leq \frac{up}{2^k}. \tag{10}$$

(b) On the other hand, if the key size of the underlying block cipher is moderate, e.g., 128 bits, then we bound the probability of the event in the following way: for a fixed key J^j, we define a following function:

$$\Sigma_{J^j}(M) := \bigoplus_{i=1}^{\ell_i - 1} \mathsf{E}_{J^j}(\langle i \rangle_s \| M[i]) \oplus (M[\ell_i] \| 10^\star).$$

Now, for a fixed message M and for a fixed arbitrary constant $c \in \{0,1\}^n$, we define the following set:

$$\mathcal{S}_c := \{j \in [g] : \Sigma_{J^j}(M) = c\}.$$

In other words, $\mathcal{S}_c$ denotes the set of ideal-cipher keys J^j such that $\Sigma_{J^j}(M) = c$. Let $\mathcal{S}$ be the set $\mathcal{S}_c$ such that the size of $\mathcal{S}_c$ is maximum over all choices of $\mathcal{S}_{c'}, c' \in \{0,1\}^n$. Therefore, we can write

$$\sum_{i=1}^{u} \Pr[\mathsf{E.1}] \quad \leq \quad \sum_{i=1}^{u} \Pr[\mathsf{E.1}, |\mathcal{S}| < \mu] + \Pr[|\mathcal{S}| \geq \mu] \tag{11}$$

We write the first term of the right-hand side of Eqn. (11) as follows:

$$\sum_{i=1}^{u} \Pr[\exists a \in [q_i], j, j' \in [g], \alpha \in [p_j] : K_2^i = J^j, \Sigma_a^i = X_\alpha^j, |\mathcal{S}| < \mu, K_1^i = J^{j'}].$$

For a fixed choice of i, a, j, j', α, the probability of the event is upper bounded by 2^{-2k} due to the randomness of $K_1^i = J^{j'}$ and $K_2^i = J^j$. However, the number of choices of (i, a) is at most q, the number of choices of (j, α) is at most p, and the number of choices of j' is at most μ. Therefore, we have

$$\sum_{i=1}^{u} \Pr[\mathsf{E.1}, |\mathcal{S}| < \mu] \leq \frac{qp\mu}{2^{2k}}. \tag{12}$$

Now, it remains to bound the last term of the right-hand side of Eqn. (11). We would like to note first that the event $|\mathcal{S}| \geq \mu$ implies that at least μ many distinct ideal-cipher keys J^j are there such that all the $\Sigma_{J^j}(M)$ values collide at a fixed but arbitrary value c. It is easy to see that for a message M and for an arbitrary but fixed constant $c \in \{0,1\}^n$,

$$\Pr[\Sigma_{J^j}(M) = c] \leq \frac{2}{2^n}, \tag{13}$$

where the probability is defined over the randomness of $\mathsf{E} \leftarrow\!\!\$\ \mathsf{BC}(\mathcal{K}, \{0,1\}^n)$. The probability follows because there will be at least one input-output for which either the input (in case of inverse ideal-cipher query) or the output (in case of forward ideal-cipher query) will be random. We use that randomness to bound the probability of the above event. Due to the independence of the keys $J^{j_1}, J^{j_2}, \ldots, J^{j_\mu}$, varying all possible choices of c, and by following Eqn. (13), the probability of the above event is $2^n(\frac{2}{2^n})^{\mu-1}$. Moreover, the number of ways we can choose the keys $J^{j_1}, J^{j_2}, \ldots, J^{j_\mu}$ is exactly $\binom{p}{\mu}$. Recall that $\mathcal{S}$ depends on the message M. Therefore, we vary all possible choices of message M, which is bounded by $\max\{p^\ell, 2^{(n-s)\ell}\}$. In the following, we consider two cases: (a) when the maximum number of message choices is p^ℓ and (b) when the maximum number of message choices is $2^{(n-s)\ell}$.

☐ Case A: maximum number of message choices is p^ℓ:

$$\Pr[|\mathcal{S}| \geq \mu] \leq 2^n p^\ell \binom{p}{\mu} \left(\frac{2}{2^n}\right)^{\mu-1} \leq 2^n p^{\ell+\mu} \cdot \frac{1}{\mu!} \cdot \left(\frac{2}{2^n}\right)^{\mu-1}. \tag{14}$$

By applying the Stirling approximation, we have

$$\Pr[|\mathcal{S}| \geq \mu] \leq 2^{2n} \cdot p^\ell \cdot \left(\frac{6p}{2^n \mu}\right)^{\mu} \leq \left(\frac{6p}{2^k}\right)^{2^{k-n}k\ell}, \tag{15}$$

where the last inequality follows from the fact $2^{2n}p^\ell \leq (k\ell)^{2^{k-n}k\ell}$, for $\mu = 2^{k-n}k\ell$.

☐ Case B: maximum number of message choices is $2^{(n-s)\ell}$:

$$\Pr[|\mathcal{S}| \geq \mu] \leq 2^n 2^{(n-s)\ell} \binom{p}{\mu} \left(\frac{2}{2^n}\right)^{\mu-1} \leq 2^n 2^{(n-s)\ell} \cdot p^\mu \cdot \frac{1}{\mu!} \cdot \left(\frac{2}{2^n}\right)^{\mu-1}. \tag{16}$$

By applying the Stirling approximation, we have

$$\Pr[|\mathcal{S}| \geq \mu] \leq 2^{2n+(n-s)\ell} \cdot \left(\frac{6p}{2^n \mu}\right)^{\mu} \leq \left(\frac{6p}{2^k}\right)^{2^{k-n}k\ell}, \tag{17}$$

where the last inequality follows from the simple algebraic calculation: $2^{2n+(n-s)\ell} \leq (k\ell)^{2^{k-n}k\ell}$, for any $k \geq 4, \ell \geq 1, s \geq 1$ and $\mu = 2^{k-n}k\ell$. Note that we make the standard assumption that $k \geq n$.

Combining Eqn. (11), Eqn. (12), Eqn. (15) and Eqn. (17) and putting $\mu = 2^{k-n}\ell k$, we have

$$\sum_{i=1}^{u} \Pr[\mathsf{E.1}] \leq \frac{pq\mu}{2^{2k}} + 2\left(\frac{6p}{2^k}\right)^{2^{k-n}k\ell} \leq \frac{pq\ell k}{2^{n+k}} + 2\left(\frac{6p}{2^k}\right)^{2^{k-n}k\ell}. \tag{18}$$

Now, we upper bound the probability of the event $\mathsf{E.2}$.

B. Bounding Event $\mathsf{E.2}$. The event $\mathsf{E.2}$ implies that the block cipher key K_1^i of the i-th user does not collide with any chosen ideal-cipher keys. As a result, the V values for computing Σ_a^i are not pre-determined. Hence, to upper bound the probability that $\Sigma_a^i = X_\alpha^j$, we are allowed to use the randomness of V variables, which in turn allows us to apply Lemma 4 to upper bound the probability of $\Sigma_a^i = X_\alpha^j$ for a fixed choice of i, a, j, and α. Moreover, the event $K_2^i = J^j$ is independent over $\Sigma_a^i = X_\alpha^j$ as the former event is based on the randomness of K_2^i and the latter one is based on the randomness of K_1^i. Therefore, for a fixed choice of indices, the probability that $K_2^i = J^j, \Sigma_a^i = X_\alpha^j$ is at most $2/2^{(n+k)}$. Therefore, by varying over all possible choices of indices, we have

$$\sum_{i=1}^{u} \Pr[\mathsf{E.2}] \leq \frac{2qp}{2^{n+k}}, \tag{19}$$

as the total number of choices of (i, a) is at most q and the total number of choices of (j, α) is at most p. Finally, by combining Eqn. (10), Eqn. (18) and Eqn. (19), and by assuming $q \leq 2^n, p \leq 2^k$, we obtain

$$\Pr[\mathsf{Bad2}] \leq \min\{\frac{up}{2^k}, \frac{q\ell k}{2^n}\} + \frac{2p}{2^k} + 2\left(\frac{6p}{2^k}\right)^{2^{k-n}k\ell} \leq \min\{\frac{up}{2^k}, \frac{q\ell k}{2^n}\} + \frac{14p}{2^k}, \tag{20}$$

where the last inequality holds as $\left(6p/2^k\right)^{2^{k-n}k\ell} \leq 6p/2^k$.

VII. Bounding $\mathsf{Bad3}$. Bounding $\mathsf{Bad3}$ is similar to that of $\mathsf{Bad2}$. Recall that the event $\mathsf{Bad3}$ holds if there exists a user i such that two of its Σ values have collided. To bound this event, for each $i \in [u]$, we define an auxiliary event as follows:

$$\mathsf{Bad}_{\mathsf{Aux}_i} \coloneqq \exists j' \in [g] : K_1^i = J^{j'}.$$

Due to the theorem of total probability, we write $\mathsf{Bad3}$ as follows:

$$\begin{aligned} \Pr[\mathsf{Bad3}] \quad &\leq \quad \sum_{i=1}^{u} \left(\Pr\left[\underbrace{\exists a \neq b \in [q_i] : \Sigma_a^i = \Sigma_b^i, \mathsf{Bad}_{\mathsf{Aux}_i}}_{\mathsf{E.1}} \right] \right) \\ &\quad + \quad \sum_{i=1}^{u} \left(\Pr\left[\underbrace{\exists a \neq b \in [q_i] : \Sigma_a^i = \Sigma_b^i, \overline{\mathsf{Bad}_{\mathsf{Aux}_i}}}_{\mathsf{E.2}} \right] \right). \end{aligned} \tag{21}$$

A. Bounding Event $\mathsf{E.1}$. The event $\mathsf{E.1}$ implies that the block cipher key K_1^i of the i-th user collides with some chosen ideal-cipher key $J^{j'}$. As a result, all the V values for computing Σ_a^i and Σ_b^i may be pre-determined, as the distinguisher can make appropriate ideal-cipher queries to determine all the V values. Hence, to upper bound the probability that $\Sigma_a^i = \Sigma_b^i$, we cannot use the randomness of V variables, which in turn prohibits us from applying Lemma 4 to upper bound the probability of $\Sigma_a^i = \Sigma_b^i$, for a fixed choice of i, a, and b. Therefore, we have to use the event that $K^i = J^{j'}$, for some $j' \in [g]$. Now, we bound the probability of the event $\mathsf{E.1}$ in three different ways as follows:

(a) Method-I: We bound the event $\Sigma_a^i = \Sigma_b^i, K_1^i = J^{j'}$ to at most 2^{-k}, by using the randomness of the event $K_1^i = J^{j'}$. However, by varying all possible choices of indices, we have roughly $up/2^k$ bound. Therefore, we have

$$\sum_{i=1}^{u} \Pr\left[\exists a \neq b \in [q_i] : \Sigma_a^i = \Sigma_b^i, \mathsf{Bad}_{\mathsf{Aux}_i}\right] \leq \frac{up}{2^k}. \tag{22}$$

(b) Method-II: It is easy to see that for two distinct messages,

$$\Pr[\Sigma_{J^j}(M) = \Sigma_{J^j}(M')] \leq \frac{2}{2^n}, \tag{23}$$

where the probability is defined over the randomness of $\mathsf{E} \leftarrow\!\$\ \mathsf{BC}(\mathcal{K}, \{0,1\}^n)$ and the probability follow because there will be at least one input-output for which either the input (in case of inverse ideal-cipher query) or the output (in case of forward ideal-cipher query) will be random. We use that randomness to bound the probability of the above event. Now, for a fixed pairs of messages M and M', we define the following set:

$$\mathcal{S} := \{j \in [g] : \Sigma_{J^j}(M) = \Sigma_{J^j}(M')\}.$$

Therefore, we can write

$$\sum_{i=1}^{u} \Pr[\mathsf{E.1}] \quad \leq \quad \sum_{i=1}^{u} \Pr[\mathsf{E.1}, |\mathcal{S}| < \mu] + \Pr[|\mathcal{S}| \geq \mu]. \tag{24}$$

We write the first term of the right hand side of Eqn. (24) as follows:

$$\sum_{i=1}^{u} \Pr\left[\exists a \neq b \in [q_i] : \Sigma_a^i = \Sigma_b^i, |\mathcal{S}| < \mu, \mathsf{Bad}_{\mathsf{Aux}_i}\right].$$

For a fixed choice of i, a, b, the probability of the event is upper bounded by 2^{-k} due to the randomness of $K_1^i = J^{j'}$. However, the number of choices of (i, a, b) is at most $q_{\max} q$ and the number of choices of j' is at most μ. Therefore, we have

$$\sum_{i=1}^{u} \Pr[\mathsf{E.1}, |\mathcal{S}| < \mu] \leq \frac{q_{\max} q \mu}{2^k}. \tag{25}$$

Now, it remains to bound the last term of the right-hand side of Eqn. (24). Using a similar technique, we have the following:

$$\Pr[|\mathcal{S}| \geq \mu] \leq 2\left(\frac{6p}{2^k}\right)^{2^{k-n} k\ell}. \tag{26}$$

By combining Eqn. (24), Eqn. (25), and Eqn. (26), we have

$$\sum_{i=1}^{u} \Pr[\mathsf{E.1}] \leq \frac{q_{\max} q \mu}{2^k} + 2\left(\frac{6p}{2^k}\right)^{2^{k-n} k\ell} \tag{27}$$

(c) Method-III: We bound the event $\mathsf{E.1}$ in two cases as follows: (a) when $i < \sqrt{q}$ and (b) when $i \geq \sqrt{q}$, i.e., we write

$$\begin{aligned}
\sum_{i=1}^{u} \Pr[\mathsf{E.1}] &= \sum_{i=1}^{\sqrt{q}-1} \Pr[\mathsf{E.1}] + \sum_{i=\sqrt{q}}^{u} \Pr[\mathsf{E.1}] \\
&\leq \sum_{i=1}^{\sqrt{q}-1} \Pr[\exists j' \in [g] : K_1^i = J^{j'}] + \sum_{i=\sqrt{q}}^{u} \sum_{a,b=1}^{q_i} \Pr[\exists j' \in [g] : \Sigma_a^i = \Sigma_b^i, K_1^i = J^{j'}] \\
&\leq \frac{p\sqrt{q}}{2^k} + \sum_{i=\sqrt{q}}^{u} \sum_{a,b=1}^{q_i} \Pr[\underbrace{\exists j' \in [g] : \Sigma_a^i = \Sigma_b^i, K_1^i = J^{j'}}_{\mathsf{E}}],
\end{aligned} \tag{28}$$

where recall that $\mathsf{Bad}_{\mathsf{Aux}_i}$ denotes the event that there exists an $j' \in [g]$ such that $K_1^i = J^{j'}$.

Now, to bound the probability of the event E, we define a set $\mathcal{S}$ for a fixed pair of messages M, M' as follows:

$$\mathcal{S} \coloneqq \{j \in [g] : \Sigma_{J^j}(M) = \Sigma_{J^j}(M')\} .$$

Therefore, we can write

$$\sum_{i=\sqrt{q}}^{u} \sum_{a,b=1}^{q_i} \Pr[\mathsf{E}] \quad = \quad \sum_{i=\sqrt{q}}^{u} \sum_{a,b=1}^{q_i} \Pr[\mathsf{E}, |\mathcal{S}| < \mu] + \Pr[|\mathcal{S}| \geq \mu]. \tag{29}$$

By combining Eqn. (28) and Eqn. (29), we have

$$\sum_{i=1}^{u} \Pr[\mathsf{E.1}] \quad \leq \quad \frac{p\sqrt{q}}{2^k} + \sum_{i=\sqrt{q}}^{u} \frac{q_i^2 \mu}{2^k} + \Pr[|\mathcal{S}| \geq \mu] \overset{(1)}{\leq} \frac{p\sqrt{q}}{2^k} + \frac{q^{3/2}\mu}{2^k} + 2\left(\frac{6p}{2^k}\right)^{2^{k-n}k\ell}, \tag{30}$$

where inequality (1) follows since $\sum_{i\in[\sqrt{q},u]} q_i^2 \leq q^{3/2}$ by Lemma 2, and the bound on $\Pr[|\mathcal{S}| \geq \mu]$ is inherited from the previous analysis. In the proof of Lemma 2, the queries are assumed to be arranged in non-increasing order, i.e., $q_1 \geq q_2 \geq \cdots \geq q_u$. In particular, we only assume $q_i \leq q^{1/2}$ for all $i \geq \sqrt{q}$. This is consistent with the ordering: if for some $i \geq \sqrt{q}$ we had $q_i > q^{1/2}$, then necessarily $q_1, q_2, \ldots, q_{i-1} \geq q_i > q^{1/2}$, which would imply $q_1 + q_2 + \cdots + q_u > q$, a contradiction.
Finally, by combining Eqn. (22), Eqn. (27) and Eqn. (30), we have

$$\begin{aligned}
\sum_{i=1}^{u} \Pr[\mathsf{E.1}] \quad &\leq \quad \min\left\{\frac{up}{2^k}, \frac{q_{\max} q\mu}{2^k} + 2\left(\frac{6p}{2^k}\right)^{2^{k-n}k\ell}, \frac{p\sqrt{q}}{2^k} + \frac{q^{3/2}\mu}{2^k} + 2\left(\frac{6p}{2^k}\right)^{2^{k-n}k\ell}\right\} \\
&\leq \quad \min\left\{\frac{up}{2^k}, \frac{q_{\max} q\ell k}{2^n}, \frac{p\sqrt{q}}{2^k} + \frac{q^{3/2}\ell k}{2^n}\right\} + \frac{12p}{2^k},
\end{aligned} \tag{31}$$

where the last inequality follows by plugging in the value of $\mu = 2^{k-n}k\ell$ and by the inequality that $\left(6p/2^k\right)^{2^{k-n}k\ell} \leq 6p/2^k$. Now, we upper bound the probability of the event $\mathsf{E.2}$.

B. Bounding Event $\mathsf{E.2}$. The event $\mathsf{E.2}$ implies that the block cipher key K_1^i of the i-th user does not collide with any chosen ideal-cipher keys. As a result, the V values for computing Σ_a^i are not pre-determined. Hence, to upper bound the probability that $\Sigma_a^i = \Sigma_b^i$, we are allowed to use the randomness of V variables, which in turn allows us to apply Lemma 4 to upper bound the probability of $\Sigma_a^i = \Sigma_b^i$, for a fixed choice of i, a, and b. Therefore, for a fixed choice of indices, the probability that $\Sigma_a^i = \Sigma_b^i$ is at most $2/2^n$. Therefore, by varying over all possible choices of indices, we have

$$\sum_{i=1}^{u} \Pr[\mathsf{E.2}] \leq \sum_{i=1}^{u} \frac{2\binom{q_i}{2}}{2^n} \leq \sum_{i=1}^{u} \frac{q_i^2}{2^n} \leq q_{\max} \sum_{i=1}^{u} \frac{q_i}{2^n} \leq \frac{q_{\max} q}{2^n}. \tag{32}$$

Finally, combining Eqn. (31) and Eqn. (32), we obtain

$$\Pr[\mathsf{Bad3}] \leq \frac{q_{\max} q}{2^n} + \min\left\{\frac{up}{2^k}, \frac{q_{\max} q\ell k}{2^n}, \frac{p\sqrt{q}}{2^k} + \frac{q^{3/2}\ell k}{2^n}\right\} + \frac{12p}{2^k}. \tag{33}$$

Finally, by combining Eqn. (4), Eqn. (5), Eqn. (6), Eqn. (9), Eqn. (20), Eqn. (33), we obtain the bound of Lemma 5.

5.3 Analysis of Good Transcripts

In this section, we compute a lower bound for the ratio of the real to ideal interpolation probability for a good transcript τ. In particular, we show that for a good transcript $\tau = (\tau_c, \tau_p)$, realizing τ is almost as likely in the real world as in the ideal world.

Lemma 6 (Good Lemma). *Let $\tau = (\tau_c, \tau_p) \in \mathsf{GoodT}$ be a good transcript. Let X_{re} and X_{id} be defined as above. Then*

$$\frac{\Pr[\mathsf{X}_{\mathrm{re}} = \tau]}{\Pr[\mathsf{X}_{\mathrm{id}} = \tau]} \geq 1.$$

Proof. For each $i \in [u]$, we first define the following tuple:

$$\widetilde{\Sigma}^i := (\Sigma_1^i, \Sigma_2^i, \ldots, \Sigma_{q_i}^i).$$

Note that, for each $i \in [u]$, all the elements in the tuple $\widetilde{\Sigma}^i$ are distinct by the virtue of $\overline{\mathsf{Bad3}}$. Moreover, as $\overline{\mathsf{BadK1}}$ holds true, we must have $K_2^i \neq K_2^j$ and $K_1^i \neq K_1^j$, for all $i \neq j \in [u]$. Let us assume that out of the u distinct keys $(K_b^1, K_b^2, \ldots, K_b^u)$ for $b \in \{1, 2\}$, some of them collide with the chosen ideal-cipher keys $(J^1, \ldots, J^g)$. Formally, for each $b \in \{1, 2\}$, define

$$\mathcal{K}_{b,\mathsf{coll}} \triangleq \{K_b^i : \exists j \in [g],\, K_b^i = J^j\},$$

and let $\beta_b = |\mathcal{K}_{b,\mathsf{coll}}|$ denote the number of such collisions. Then, by the virtue of $\overline{\mathsf{Bad2}}$, for each $i \in [u]$ such that $K_2^i \in \mathcal{K}_{2,\mathsf{coll}}$, none of the elements of the tuple $\widetilde{\Sigma}^i$ collides with any elements of $\mathsf{Dom}(\mathsf{E}_{J^j})$, for some $j \in [g]$ such that $K_2^i = J^j$ holds, i.e., for all $a \in [q_i], \Sigma_a^i \neq X_\alpha^j$ for some $j \in [g]$ such that $K_2^i = J^j$ and for all $\alpha \in [p_j]$. Similarly, for each $i \in [u]$ such that $K_2^i \in \mathcal{K}_{2,\mathsf{coll}}$, by the virtue of $\overline{\mathsf{Bad1}}$, we have $T_a^i \neq Y_\alpha^j$, for all $a \in [q_i]$, for some $j \in [g]$ such that $K_2^i = J^j$ and for all $\alpha \in [p_j]$. Let for $b \in \{1, 2\}$, $\mathcal{J}_b = \{(j, i) \in [g] \times [u] : K_b^i = J^j\}$. Note that, due to $\overline{\mathsf{BadK1}}$, we have $|\mathcal{J}_b| = |\mathcal{K}_{b,\mathsf{coll}}| = \beta_b$. For each $b \in \{1, 2\}$ we write $\mathcal{J}_b^*$ to denote the set of all $j \in [g]$ such that $(j, \star) \in \mathcal{J}_b$. For $(j, i) \in \mathcal{J}_1$, let the block cipher used in the hash part be invoked a total of $\zeta_{i,j}$ times, including ideal cipher queries and message block queries. Moreover, assume $\sigma^i[\alpha]$ denotes the total number of distinct message blocks at the α-th position across all q_i queries, where $\alpha \in \ell^i$, and ℓ^i denotes the maximum number of message blocks queried across all q_i queries for the i-th user. Let

$$\sigma^i = \sum_{\alpha=1}^{\ell^i} \sigma^i[\alpha].$$

Now, we compute the ideal and real interpolation probability for a good transcript as follows.

<u>Ideal Interpolation Probability:</u> Note that, in the online phase of the game, the ideal world samples the response T_a^i independently and uniformly at random for each query M_a^i. Moreover, in the offline phase, it uniformly and independently samples u pairs of keys and also computes $V_a^i[\alpha]$ for all $i \in [u], a \in [q_i]$ and $\alpha \in [\ell_a^i]$. Recall that for $(j, i) \in \mathcal{J}_1$, the block cipher in the hash part is invoked $\zeta_{i,j}$ times in total. Therefore, we have

$$\Pr[\mathsf{X}_{\mathrm{id}} = \tau] = \prod_{i=1}^{u} \frac{1}{2^{nq_i}} \cdot \prod_{(j,i) \in \mathcal{J}_1} \frac{1}{\mathbf{P}(2^n, \zeta_{i,j})} \cdot \prod_{\substack{i \in [u] \\ K_1^i \notin \mathcal{K}_{1,\mathsf{coll}}}} \frac{1}{\mathbf{P}(2^n, \sigma^i)} \cdot$$

$$\cdot \prod_{j \in [g] \setminus \mathcal{J}_1^*} \frac{1}{\mathbf{P}(2^n, p_j)} \cdot \prod_{i=1}^{u} \frac{1}{2^{2k}}.$$

<u>Real Interpolation Probability:</u> To compute the real interpolation probability, first of all, we count the total number of times the block cipher has been invoked with different keys to compute Σ values. It is easy to see that for the i-th user, the total number of block ciphers called to compute $\widetilde{\Sigma}^i$ tuple is σ^i. Moreover, for each $(j,i) \in \mathcal{J}_2$, the block cipher used in the finalization phase is invoked for a total of $p_j + q_i$ times, and for all those $i \in [u]$ such that $K_2^i \notin \mathcal{K}_{2,\mathsf{coll}}$, the block cipher in the finalization phase is invoked for a total of q_i times with the user key K_2^i. Besides, as the transcript is good, both K_1^i and K_2^i are distinct for each $i \in [u]$. Therefore, we have

$$\Pr[\mathsf{X}_{\mathrm{re}} = \tau] = \prod_{(j,i)\in\mathcal{J}_2} \frac{1}{\mathbf{P}(2^n, p_j + q_i)} \cdot \prod_{\substack{i\in[u]\\ K_2^i\notin\mathcal{K}_{2,\mathsf{coll}}}} \frac{1}{\mathbf{P}(2^n, q_i)} \cdot \prod_{(j,i)\in\mathcal{J}_1} \frac{1}{\mathbf{P}(2^n, \zeta_{i,j})} \cdot \prod_{\substack{i\in[u]\\ K_1^i\notin\mathcal{K}_{1,\mathsf{coll}}}} \frac{1}{\mathbf{P}(2^n, \sigma^i)} \cdot \prod_{j\in[g]\setminus\{\mathcal{J}_1^*,\mathcal{J}_2^*\}} \frac{1}{\mathbf{P}(2^n, p_j)} \cdot \prod_{i=1}^{u} \frac{1}{2^{2k}}.$$

Note that, by rearranging terms, we have

$$\begin{aligned}
\Pr[\mathsf{X}_{\mathrm{re}} = \tau] &= \prod_{(j,i)\in\mathcal{J}_2} \frac{1}{\mathbf{P}(2^n, p_j + q_i)} \cdot \prod_{\substack{i\in[u]\\ K_2^i\notin\mathcal{K}_{2,\mathsf{coll}}}} \frac{1}{\mathbf{P}(2^n, q_i)} \cdot \prod_{(j,i)\in\mathcal{J}_1} \frac{1}{\mathbf{P}(2^n, \zeta_{i,j})} \\
&\quad \cdot \prod_{\substack{i\in[u]\\ K_1^i\notin\mathcal{K}_{1,\mathsf{coll}}}} \frac{1}{\mathbf{P}(2^n, \sigma^i)} \cdot \prod_{j\in[g]\setminus\{\mathcal{J}_1^*,\mathcal{J}_2^*\}} \frac{1}{\mathbf{P}(2^n, p_j)} \cdot \prod_{i=1}^{u} \frac{1}{2^{2k}} \\
&= \prod_{(j,i)\in\mathcal{J}_2} \frac{1}{\mathbf{P}(2^n, p_j)} \cdot \prod_{(j,i)\in\mathcal{J}_2} \frac{1}{\mathbf{P}(2^n - p_j, q_i)} \cdot \prod_{\substack{i\in[u]\\ K_2^i\notin\mathcal{K}_{2,\mathsf{coll}}}} \frac{1}{\mathbf{P}(2^n, q_i)} \\
&\quad \cdot \prod_{(j,i)\in\mathcal{J}_1} \frac{1}{\mathbf{P}(2^n, \zeta_{i,j})} \cdot \prod_{\substack{i\in[u]\\ K_1^i\notin\mathcal{K}_{1,\mathsf{coll}}}} \frac{1}{\mathbf{P}(2^n, \sigma^i)} \cdot \prod_{j\in[g]\setminus\{\mathcal{J}_1^*,\mathcal{J}_2^*\}} \frac{1}{\mathbf{P}(2^n, p_j)} \cdot \prod_{i=1}^{u} \frac{1}{2^{2k}} \\
&\overset{(1)}{=} \prod_{(j,i)\in\mathcal{J}_2} \frac{1}{\mathbf{P}(2^n - p_j, q_i)} \cdot \prod_{\substack{i\in[u]\\ K_2^i\notin\mathcal{K}_{2,\mathsf{coll}}}} \frac{1}{\mathbf{P}(2^n, q_i)} \cdot \prod_{(j,i)\in\mathcal{J}_1} \frac{1}{\mathbf{P}(2^n, \zeta_{i,j})} \\
&\quad \cdot \prod_{\substack{i\in[u]\\ K_1^i\notin\mathcal{K}_{1,\mathsf{coll}}}} \frac{1}{\mathbf{P}(2^n, \sigma^i)} \cdot \prod_{j\in[g]\setminus\mathcal{J}_1^*} \frac{1}{\mathbf{P}(2^n, p_j)} \cdot \prod_{i=1}^{u} \frac{1}{2^{2k}},
\end{aligned}$$

where (1) follows from the fact that $(j,i) \in \mathcal{J}_2 \iff j \in \mathcal{J}_2^*$. Now, by taking the ratio of the real to ideal interpolation probability, we have

$$\frac{\Pr[\mathsf{X}_{\mathrm{re}} = \tau]}{\Pr[\mathsf{X}_{\mathrm{id}} = \tau]} = \prod_{i\in[u]:K_2^i\notin\mathcal{K}_{2,\mathsf{coll}}} \frac{2^{nq_i}}{\mathbf{P}(2^n, q_i)} \cdot \prod_{(j,i)\in\mathcal{J}_2} \frac{2^{nq_i}}{\mathbf{P}(2^n - p_j, q_i)} \geq 1,$$

which finally proves Lemma 6. Finally, by combining Lemma 5 and Lemma 6 in the H-Coefficient framework, we obtain the desired security bound of $\mathsf{LightMAC}$ construction in the multi-user setting.

6 Proof of Theorem 5

We consider a computationally unbounded non-trivial deterministic distinguisher A that is given access to u independent instances of the $\mathsf{LightMAC\_Plus}$ construction, denoted as

$(\mathsf{LightMAC\_Plus}[\mathsf{E}]_{(K_1^i,K_2^i,K_3^i)})_{i\in[u]}$ in the real world, where each (K_1^i, K_2^i, K_3^i) is independent of (K_1^j, K_2^j, K_3^j) and $\mathsf{E} \leftarrow\!\!\$\ \mathsf{BC}(\mathcal{K}, \{0,1\}^n)$ is an ideal block cipher. In the ideal world, it has access to a tuple of u independent random functions $(\mathsf{RF}_1, \ldots, \mathsf{RF}_u)$. In both worlds, A is given additional access to the oracle $\mathsf{E}^{\pm}$. In the real world, E is the block cipher underneath the construction $\mathsf{LightMAC\_Plus}$ and in the ideal world E is uniformly sampled from $\mathsf{BC}(\mathcal{K}, \{0,1\}^n)$ independent of the distribution of the sequence of u independent random functions.

The interaction between the distinguisher and the challenger is summarized in a transcript $\tau_c = \tau_c^1 \cup \tau_c^2 \cup \ldots \cup \tau_c^u$, where $\tau_c^i = \{(M_1^i, T_1^i), \ldots, (M_{q_i}^i, T_{q_i}^i)\}$ and the ideal-cipher queries are summarized in a transcript $\tau_p = \tau_p^1 \cup \tau_p^2 \cup \ldots \cup \tau_p^g$, where $\tau_p^j = \{(u_1^j, v_1^j), \ldots, (u_{p_j}^j, v_{p_j}^j), J^j\}$. As before, A makes q_i construction queries and p_j ideal cipher queries (including forward and inverse queries) with chosen ideal cipher key J^j. Let $q = (q_1 + \ldots + q_u)$ be the total number of construction queries $p = (p_1 + \ldots + p_g)$ be the total number of ideal cipher queries. We modify the experiment by releasing internal information to A after it has finished its interaction but has not yet output the decision bit. In the real world, we reveal all the keys (K_1^i, K_2^i, K_3^i) for all u users and also the $(V_a^i[\alpha])_{i\in[q],a\in[q_i],\alpha\in[\ell_a^i]}$ tuple, where ℓ_a^i is the maximum number of message blocks corresponding the a-th query of the i-th user and $V_a^i[\alpha] = \mathsf{E}_{K_1^i}(\langle\alpha\rangle_s \| M_a^i[\alpha])$. In the ideal world, the challenger sample the keys (K_1^i, K_2^i, K_3^i) uniformly at random from their respective key spaces for all u users and computes the tuple $(V_a^i)_{i\in[q],a\in[q_i]}$ tuple as follows:

$$V_a^i[\alpha] = \mathsf{E}_{K_1^i}(\langle\alpha\rangle_s \| M_a^i[\alpha])$$

and reveal them to the distinguisher. Therefore, each transcript τ_i^c, where $i \in [u]$, is now modified to include the corresponding triplet of keys and the $V_a^i[\alpha]$ values for the i-th instance of the construction. Thus, the modified transcript is

$$\tilde{\tau}_c^i = \{(M_1^i, T_1^i, \widetilde{V}_1^i), \ldots, (M_{q_i}^i, T_{q_i}^i, \widetilde{V}_{q_i}^i), (K_1^i, K_2^i, K_3^i)\},$$

where $\widetilde{V}_a^i := (V_a^i[\alpha])_{\alpha\in[\ell_a^i]}$. In all the following, the complete construction query transcript is $\tau_c = \bigcup_{i=1}^{u} \tilde{\tau}_c^i$ and the overall transcript is $\tau = \tau_c \cup \tau_p$. The modified experiment only makes the distinguisher more powerful, and hence the distinguishing advantage of A in this experiment is no less than its distinguishing advantage in the former. Therefore, it follows that for each $i \in [u]$, $\mathsf{LightMAC\_Plus}[\mathsf{E}]_{(K_1^i,K_2^i,K_3^i)} \mapsto \tilde{\tau}_c^i$ denotes the following:

(a) $\Sigma_a^i = \bigoplus_{\alpha=1}^{\ell_a^i} V_a^i[\alpha]$, $a \in [q_i]$

(b) $\Theta_a^i = \bigoplus_{\alpha=1}^{\ell_a^i} 2^{\ell_a^i-\alpha} V_a^i[\alpha]$, $a \in [q_i]$

(c) $\mathsf{E}_{K_2^i}(\Sigma_a^i) \oplus \mathsf{E}_{K_3^i}(\Theta_a^i) = T_a^i$, $a \in [q_i]$.

6.1 Definition and Bounding Probability of Bad Transcripts

In this section, we define and bound the probability of bad transcripts in the ideal world. We denote the input to the α-th ideal-cipher query corresponding to the ideal-cipher key J^j as X_α^j and the corresponding response as Y_α^j. We say that an attainable transcript $\tau = (\tau_c, \tau_p)$ is a **bad** transcript if any one of the following events holds:

1. **Bad Events Based on Key Collision:**

1. BadK1 : $\exists i_1 \neq i_2 \in [u]$ such that $K_1^{i_1} = K_1^{i_2}, K_2^{i_1} = K_2^{i_2}$.
2. BadK2 : $\exists i_1 \neq i_2 \in [u]$ such that $K_1^{i_1} = K_1^{i_2}, K_3^{i_1} = K_3^{i_2}$.
3. BadK3 : $\exists i_1 \neq i_2, i_3 \in [u]$ such that $K_2^{i_1} = K_2^{i_2}, K_3^{i_1} = K_3^{i_3}$.
4. BadK4 : $\exists i_1 \neq i_2 \in [u], j \in [g], j' \in [g], b \in \{2, 3\}$ such that $K_b^{i_1} = K_b^{i_2}, K_1^{i_1} = J^j, K_1^{i_2} = J^{j'}$.
5. BadK5 : $\exists i \in [u], j, j' \in [g]$ such that $K_2^i = J^j, K_3^i = J^{j'}$.
6. BadK6 : $\exists i \in [u]$ such that $K_2^i = K_3^i$.
7. BadK7 : $\exists i_1 \neq i_2 \in [u], a \in [q_{i_1}], b \in [q_{i_2}], m \in [l_a^{i_1}]$ such that $K_1^{i_1} = K_2^{i_2}, \langle m \rangle \| M_a^{i_1}[m] = \Sigma_b^{i_2}$.
8. BadK8 : $\exists i_1 \neq i_2 \in [u], a \in [q_{i_1}], b \in [q_{i_2}], m \in [l_a^{i_1}]$ such that $K_1^{i_1} = K_3^{i_2}, \langle m \rangle \| M_a^{i_1}[m] = \Theta_b^{i_2}$.

2. **Bad Events Based on Input and Key Collision:**

1. Bad1 : $\exists i \in [u], a \in [q_i], j \in [g], \alpha \in [p_j]$ such that $K_2^i = J^j, \Sigma_a^i = X_\alpha^j$.
2. Bad2 : $\exists i \in [u], a \in [q_i], j \in [g], \alpha \in [p_j]$ such that $K_3^i = J^j, \Theta_a^i = X_\alpha^j$.
3. Bad3 : $\exists i \in [u], a \neq b \in [q_i], j \in [g]$ such that $\Sigma_a^i = \Sigma_b^i, K_1^i = J^j$.
4. Bad4 : $\exists i \in [u], a \neq b \in [q_i], j \in [g]$ such that $\Theta_a^i = \Theta_b^i, K_1^i = J^j$.
5. Bad5 : $\exists i \in [u], a \neq b \in [q_i]$ such that $\Sigma_a^i = \Sigma_b^i, T_a^i = T_b^i$.
6. Bad6 : $\exists i \in [u], a \neq b \in [q_i]$ such that $\Theta_a^i = \Theta_b^i, T_a^i = T_b^i$.
7. Bad7 : $\exists i_1 \neq i_2 \in [u], a \in [q_{i_1}], b \in [q_{i_2}]$ such that $K_2^{i_1} = K_2^{i_2}, \Sigma_a^{i_1} = \Sigma_b^{i_2}$.
8. Bad8 : $\exists i_1 \neq i_2 \in [u], a \in [q_{i_1}], b \in [q_{i_2}]$ such that $K_2^{i_1} = K_2^{i_2}, \Theta_a^{i_1} = \Theta_b^{i_2}$.
9. Bad9 : $\exists i_1 \neq i_2 \in [u], a \in [q_{i_1}], b \in [q_{i_2}]$ such that $K_3^{i_1} = K_3^{i_2}, \Sigma_a^{i_1} = \Sigma_b^{i_2}$.
10. Bad10 : $\exists i_1 \neq i_2 \in [u], a \in [q_{i_1}], b \in [q_{i_2}]$ such that $K_3^{i_1} = K_3^{i_2}, \Theta_a^{i_1} = \Theta_b^{i_2}$.
11. Bad11 : $\exists i \in [u], a, b \in [q_i]$ such that $\Sigma_a^i = \Sigma_b^i, \Theta_a^i = \Theta_b^i$.
12. Bad12 : $\exists i \in [u], a, b, c \in [q_i]$ such that $\Sigma_a^i = \Sigma_b^i, \Theta_a^i = \Theta_c^i$.
13. Bad13 : $|\{(a, b) : \Sigma_a^i = \Sigma_b^i\}| \geq q_i^{1/2}$.
14. Bad14 : $|\{(a, b) : \Theta_a^i = \Theta_b^i\}| \geq q_i^{1/2}$.

Recall that $\mathsf{BadT} \subseteq \mathcal{V}$ is the set of all attainable bad transcripts and $\mathsf{GoodT} = \mathcal{V} \setminus \mathsf{BadT}$ is the set of all attainable good transcripts. Before we proceed to bound the above events in the ideal world, we state the following lemma from [Nai17, Lemma 1] that upper bounds the simultaneous collision probability between two Σ values and Θ values. We emphasize that the following results will be frequently used in upper-bounding the probability of the above bad events.

Lemma 7 (**Collision of** (Σ, Θ))**.** *For three distinct messages M_a, M_b and M_c, we have*

$$\Pr[\Sigma_a = \Sigma_b, \Theta_a = \Theta_c] \leq \frac{4}{2^{2n}},$$

provided the maximum number of message blocks $\ell \leq 2^{n/2}$. [3]

For the subsequent analysis, we also need to bound the collision probability of Θ, which is formally captured in the following lemma.

Lemma 8 (**Collision of** Θ)**.** *Let M_a and M_b be two distinct messages such that the maximum number of message blocks is ℓ. Then, we have*

$$\Pr[\Theta_a = \Theta_b] \leq \frac{2}{2^n},$$

provided the maximum number of message blocks $\ell \leq 2^{n/2}$.

Proof of the above lemma is similar to that of Lemma 4 and thus can be followed from [LPTY16, Proposition 1]. The following lemma upper bounds the probability of realizing a bad transcript in the ideal world.

Lemma 9 (**Bad Lemma**)**.** *Let $\tau = (\tau_c, \tau_p)$ be any attainable transcript. Let X_{id} and BadT be defined as above. Then, by assuming $u \leq 2^k$, we have*

$$\begin{aligned} \Pr[\mathsf{X}_{\mathrm{id}} \in \mathsf{BadT}] \quad \leq \quad & \frac{2u}{2^k} + \frac{u^3}{6.2^{2k}} + \frac{2up^2}{2^{2k}} + \frac{6q^2\ell}{2^{n+k}} + \frac{5qq_{\max}^2}{2^{2n}} + \frac{2p\sqrt{q}}{2^k} \\ & + \frac{2q^{3/2}k\ell}{2^n} + \frac{2q\sqrt{q_{\max}}}{2^n} + \min\{\frac{2up}{2^k}, \frac{2q\ell k}{2^n}\} + \frac{52p}{2^k}. \end{aligned}$$

Proof of the lemma is deferred to Appendix A.

6.2 Analysis of Good Transcripts

In this section, we compute a lower bound for the ratio of the real to ideal interpolation probability for a good transcript τ. Let us define the following sets for each $b \in \{2, 3\}$.

$$\mathcal{I}_b^{=} \quad \stackrel{\Delta}{=} \quad \{i \in [u] : \exists j \in [g], K_b^i = J^j\}.$$

We define an equivalence relation $\sim_b$ for $b \in \{2, 3\}$, over $\mathcal{I}_b^{=}$ as follows: $i_1 \sim_b i_2$ if and only if $K_b^{i_1} = K_b^{i_2}$. As a result, $\sim_b$ induces a partition over the set $\mathcal{I}_b^{=}$ as follows:

$$\mathcal{I}_b^{=} = \mathcal{I}_b^{=}[1] \sqcup \ldots \sqcup \mathcal{I}_b^{=}[r_b].$$

In other words, $\mathcal{I}_b^{=}[j]$ is the set of all users i such that $K_b^i = J^j$. It is easy to see that due to $\overline{\mathsf{BadK5}}$, $\mathcal{I}_2^{=} \cap \mathcal{I}_3^{=} = \emptyset$. We define a set $\mathcal{U}^{=}$ as follows:

$$\mathcal{U}^{=} := \{i \in [u] : i \in \mathcal{I}_2^{=} \cup \mathcal{I}_3^{=}\}.$$

For each $j \in [r_b]$ and for each $i \in \mathcal{I}_b^{=}[j]$, we consider the sequences

$$\widetilde{\Sigma}^i := (\Sigma_1^i, \Sigma_2^i, \ldots, \Sigma_{q_i}^i), \widetilde{\Theta}^i := (\Theta_1^i, \Theta_2^i, \ldots, \Theta_{q_i}^i).$$

From these sequences, we construct a bipartite graph G_i^b, where the nodes in one partition represent Σ_a^i values and the nodes in the other represent Θ_a^i values, for $i \in \mathcal{I}_b^{=}$. We put an edge between the node corresponding to Σ_a^i and Θ_a^i with the label T_a^i if $\Sigma_a^i \oplus \Theta_a^i = T_a^i$

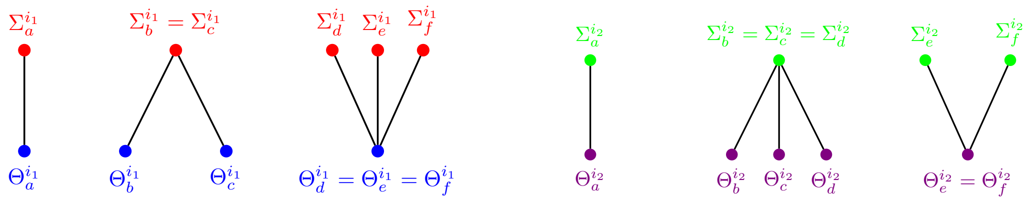

Figure 7: Distinct tuples for users i_1 and i_2 in $\mathcal{I}_b^{=}[j]$: Elements within $\widetilde{\Sigma}^{i_1}$ differ from those in $\widetilde{\Sigma}^{i_2}$, and likewise, elements within $\widetilde{\Theta}^{i_1}$ from $\widetilde{\Theta}^{i_2}$, due to conditions $\overline{\mathsf{Bad7}}$ through $\overline{\mathsf{Bad10}}$.

holds. For $i \in \mathcal{I}_b^{=}$, if $\Sigma_a^i = \Sigma_b^i$ or $\Theta_a^i = \Theta_b^i$, then we merge the corresponding nodes into a single node.
Note that, for each $i \in \mathcal{I}_b^{=}[j]$, if $\Sigma_a^i = \Sigma_b^i$ (resp. $\Theta_a^i = \Theta_b^i$), then all the elements in the tuple $\widetilde{\Theta}^i$ (resp. $\widetilde{\Sigma}^i$) are distinct (due to $\overline{\mathsf{Bad11}} \wedge \overline{\mathsf{Bad12}}$). Moreover, by the virtue of $\overline{\mathsf{Bad7}} \wedge \overline{\mathsf{Bad8}}$, we have $\widetilde{\Sigma}^{i_1} \cap \widetilde{\Sigma}^{i_2} = \emptyset$ and $\widetilde{\Theta}^{i_1} \cap \widetilde{\Theta}^{i_2} = \emptyset$ for $i_1, i_2 \in \mathcal{I}_2^{=}[j]$. In a similar way, by the virtue of $\overline{\mathsf{Bad9}} \wedge \overline{\mathsf{Bad10}}$, we have $\widetilde{\Sigma}^{i_1} \cap \widetilde{\Sigma}^{i_2} = \emptyset$ and $\widetilde{\Theta}^{i_1} \cap \widetilde{\Theta}^{i_2} = \emptyset$ for $i_1, i_2 \in \mathcal{I}_3^{=}[j]$.

As the transcript is good, it is easy to see that each component is acyclic and is a star graph. Due to $\overline{\mathsf{Bad13}} \wedge \overline{\mathsf{Bad14}}$, the component size is restricted up to $q^{1/2}$. Moreover, due to $\overline{\mathsf{Bad1}} \wedge \overline{\mathsf{Bad2}}$, each vertex of the graph, i.e., Σ_a^i or Θ_a^i does not collide with the input of any ideal-cipher query such that the ideal-cipher key collides with the i-th user key. Hence, each vertex of the graph G_i^b does not collide with the input of any ideal-cipher query. Note that, $\overline{\mathsf{Bad5}}$ (resp. $\overline{\mathsf{Bad6}}$) ensures the fact that if Σ_a^i collides with Σ_b^i (resp. Θ_a^i collides with Θ_b^i), then T_a^i must be distinct from T_b^i. On the other hand, $\overline{\mathsf{Bad7}\text{-}\mathsf{Bad10}}$ implies that there should not be any intersection between the equation variables corresponding to two different users whose keys have been collided.

We define the following set $\mathcal{U}^{\neq} := \{i \in [u] : i \notin \mathcal{U}^{=}\}$. Now, for any one of $b \in \{2, 3\}$, we define the equivalence relation $\sim$ over the set $\mathcal{U}^{\neq}$ as follows:

$$i \sim j \text{ if and only if } K_2^i = K_2^j \text{ or } K_3^i = K_3^j.$$

This equivalence relation induces a partition on the set $\mathcal{U}^{\neq}$. Therefore, we have

$$\mathcal{U}^{\neq} = (\mathcal{I}^{\neq}[1] \sqcup \ldots \sqcup \mathcal{I}^{\neq}[r']).$$

As before, for each $j \in [r']$, and for each $i \in \mathcal{I}^{\neq}[j]$, consider the sequences

$$\widetilde{\Sigma}^i := (\Sigma_1^i, \Sigma_2^i, \ldots, \Sigma_{q_i}^i), \widetilde{\Theta}^i := (\Theta_1^i, \Theta_2^i, \ldots, \Theta_{q_i}^i).$$

For each $i \in \mathcal{I}^{\neq}$, we construct a bipartite graph H_i, one of whose partitions represents the nodes corresponding to Σ_a^i values and the other one represents the nodes corresponding to Θ_a^i values. We put an edge between the node corresponding to Σ_a^i and Θ_a^i with the label T_a^i, if $\Sigma_a^i \oplus \Theta_a^i = T_a^i$ holds. However, if two nodes represent the same values, then we merge them into a single node. Note that, due to $\overline{\mathsf{Bad7}}$- $\overline{\mathsf{Bad10}}$, we have $\widetilde{\Sigma}^{i_1} \cap \widetilde{\Sigma}^{i_2} = \emptyset$ and $\widetilde{\Theta}^{i_1} \cap \widetilde{\Theta}^{i_2} = \emptyset$ for $i_1, i_2 \in \mathcal{I}^{\neq}[j]$. Let $\sigma^i[\alpha]$ denote the total number of distinct message blocks at the α-th position across all q_i queries, where $\alpha \in [\ell^i]$, and ℓ^i denotes the maximum number of message blocks queried across all q_i queries for the i-th user. Let

$$\sigma^i = \sum_{\alpha=1}^{\ell^i} \sigma^i[\alpha].$$

[3]The bound on the maximum message length in terms of the number of blocks is $\min\{2^{n-1}, 2^s\}$. However, as we choose $s = n/2$ for achieving rate 1/2, the bound on ℓ is at most $2^{n/2}$.

Now, for a good transcript $\tau = (\tau_c, \tau_p)$, we show that realizing τ is almost as likely in the real world as in the ideal world. For the purpose, let us assume that out of the u distinct keys $(K_1^1, K_1^2, \ldots, K_1^u)$, some of them collide with the chosen ideal-cipher keys $(J^1, J^2, \ldots, J^g)$. Formally, we define

$$\mathcal{K}_{1,\mathsf{coll}} \triangleq \left\{K_1^i : \exists j \in [g],\, K_1^i = J^j\right\}.$$

Furthermore, we define

$$\mathcal{J}_1 \triangleq \left\{(j,i) \in [g] \times [u] : K_1^i = J^j\right\} \text{ and } \mathcal{J}_1^* \triangleq \{j \in [g] : (j, \star) \in \mathcal{J}_1\}.$$

Moreover, we assume, for $(j,i) \in \mathcal{J}_1$, that the block cipher used in the hash part is invoked a total of $\zeta_{i,j}$ times, including ideal cipher queries and message block queries. Now we move forward and compute the ideal and real interpolation probability for a good transcript as follows.

For calculating the ideal interpolation probability, each response to the construction query is sampled uniformly and independently. After the interaction is over, the ideal world samples three k-bit dummy keys uniformly and independently from $\{0,1\}^k$. Therefore, we have

$$\Pr[\mathsf{X}_{\mathrm{id}} = \tau] = \prod_{(j,i)\in\mathcal{J}_1} \frac{1}{\mathbf{P}(2^n, \zeta_{i,j})} \cdot \prod_{\substack{i\in[u]\\ K_1^i \notin \mathcal{K}_{1,\mathsf{coll}}}} \frac{1}{\mathbf{P}(2^n, \sigma^i)} \cdot \prod_{j\in[g]\setminus\mathcal{J}_1^*} \frac{1}{\mathbf{P}(2^n, p_j)} \cdot \tag{34}$$

$$\cdot \prod_{i=1}^{u} \frac{1}{2^{nq_i}} \cdot \prod_{i=1}^{u} \frac{1}{2^{3k}}. \tag{35}$$

For calculating the real interpolation probability, for each $j \in [r_2]$, let there are $\gamma_{2,j}$ be many vertices corresponding to all $\widetilde{\Sigma}^i$ values and $\delta_{2,j}$ be many vertices corresponding to all $\widetilde{\Theta}^i$ values in the graph $\cup_{i\in\mathcal{I}_2^=[j]} G_i^2$ for all $i \in \mathcal{I}_2^=[j]$. Similarly, for each $j \in [r_3]$, let there are $\gamma_{3,j}$ be many vertices corresponding to all $\widetilde{\Sigma}^i$ values and $\delta_{3,j}$ be many vertices corresponding to all $\widetilde{\Theta}^i$ values in the graph $\cup_{i\in\mathcal{I}_3^=[j]} G_i^3$ for all $i \in \mathcal{I}_3^=[j]$. Subsequently, for each $j \in [r']$, let there are η_j be many vertices corresponding to all $\widetilde{\Sigma}^i$ values and ξ_j be many vertices corresponding to all $\widetilde{\Theta}^i$ values in the graph $\cup_{i\in\mathcal{I}^{\neq}[j]} H_i$ for all $i \in \mathcal{I}^{\neq}[j]$. Thus, the real interpolation probability becomes

$$\begin{aligned}\Pr[\mathsf{X}_{\mathrm{re}} = \tau] = & \prod_{(j,i)\in\mathcal{J}_1} \frac{1}{\mathbf{P}(2^n, \zeta_{i,j})} \cdot \prod_{\substack{i\in[u]\\ K_1^i \notin \mathcal{K}_{1,\mathsf{coll}}}} \frac{1}{\mathbf{P}(2^n, \sigma^i)} \cdot \prod_{j\in[g]\setminus\mathcal{J}_1^*} \frac{1}{\mathbf{P}(2^n, p_j)} \cdot \\ & \cdot \left(\prod_{j=1}^{r_2} \frac{h(\cup_{i\in\mathcal{I}_2^=[j]} G_i^2)}{\mathbf{P}(2^n - p_j, \gamma_{2,j}) \cdot \mathbf{P}(2^n, \delta_{2,j})}\right) \cdot \left(\prod_{j=1}^{r_3} \frac{h(\cup_{i\in\mathcal{I}_3^=[j]} G_i^3)}{\mathbf{P}(2^n, \gamma_{3,j}) \cdot \mathbf{P}(2^n - p_j, \delta_{3,j})}\right) \\ & \cdot \left(\prod_{j=1}^{r'} \frac{h(\cup_{i\in\mathcal{I}^{\neq}[j]} H_i)}{\mathbf{P}(2^n, \eta_j) \cdot \mathbf{P}(2^n, \xi_j)}\right) \cdot \prod_{i=1}^{u} \frac{1}{2^{3k}},\end{aligned} \tag{36}$$

where $h(\cup_{i\in\mathcal{I}_b^=[j]} G_i^b)$ denotes the number of solutions to the graph $\cup_{i\in\mathcal{I}_b^=[j]} G_i^b$. Similarly, $h(\cup_{i\in\mathcal{I}^{\neq}[j]} H_i)$ denotes the number of solutions to the graph $\cup_{i\in\mathcal{I}^{\neq}[j]} H_i$. Due to the several bad conditions defined in Sect. 6.1, it is easy to see that both the graphs G_i^b for $i \in \mathcal{I}_b^=[j]$ and H_i for $i \in \mathcal{I}^{\neq}[j]$ are good.

For a fixed $j \in [r_b]$, let $\alpha_{b,j}$ denotes the number of components of the graph $\cup_{i\in\mathcal{I}_b^=[j]} G_i^b$. Let $c_{k,b,j}$ (resp. $d_{k,b,j}$) denotes the number of vertices corresponding to the Σ (resp. Θ)

values in the k-th component of the graph $\cup_{i\in\mathcal{I}_b^{=}[j]} G_i^b$, where $k \in [\alpha_{b,j}]$. Therefore, we have

$$\gamma_{b,j} = \sum_{k=1}^{\alpha_{b,j}} c_{k,b,j},\ \delta_{b,j} = \sum_{k=1}^{\alpha_{b,j}} d_{k,b,j}.$$

Moreover, following the notations of Sect. 2.4, we write

$$\rho_{m,b,j}[1] = \sum_{k=1}^{m} c_{k,b,j},\ \rho_{m,b,j}[2] = \sum_{k=1}^{m} d_{k,b,j},$$

where $m \in [\alpha_{b,j}]$. Therefore, by applying Lemma 3, we have the following bound of $h(\cup_{i\in\mathcal{I}_b^{=}[j]} G_i^b)$ for a fixed j and $b \in \{2,3\}$:

$$\begin{aligned}
h(\cup_{i\in\mathcal{I}_2^{=}[j]} G_i^2) \quad &\geq \quad \frac{\mathbf{P}(2^n - p_j, \gamma_{2,j}) \cdot \mathbf{P}(2^n, \delta_{2,j})}{2^{n\sum\limits_{i\in\mathcal{I}_2^{=}[j]} q_i}} \Bigg(1 - \frac{1}{2^{2n}} \cdot \Bigg(\sum_{k=1}^{\alpha_{2,j}} 10 \Bigg(\binom{c_{k,2,j} + d_{k,2,j}}{2} \\
&\qquad \Big(p_j + \rho_{k-1,2,j}[1] + \rho_{k-1,2,j}[2]\Big)^2\Bigg)\Bigg)\Bigg), \\
&\overset{(1)}{\geq} \quad \frac{\mathbf{P}(2^n - p_j, \gamma_{2,j}) \cdot \mathbf{P}(2^n, \delta_{2,j})}{2^{n\sum\limits_{i\in\mathcal{I}_2^{=}[j]} q_i}} \Bigg(1 - \frac{1}{2^{2n}} \cdot \Bigg(\sum_{k=1}^{\alpha_{2,j}} 10 \Bigg(\binom{c_{k,2,j} + d_{k,2,j}}{2} \\
&\qquad \Big(p_j + 4q_{2,j}\Big)^2\Bigg)\Bigg)\Bigg), \\
h(\cup_{i\in\mathcal{I}_3^{=}[j]} G_i^3) \quad &\geq \quad \frac{\mathbf{P}(2^n, \gamma_{3,j}) \cdot \mathbf{P}(2^n - p_j, \delta_{3,j})}{2^{n\sum\limits_{i\in\mathcal{I}_3^{=}[j]} q_i}} \Bigg(1 - \frac{1}{2^{2n}} \cdot \Bigg(\sum_{k=1}^{\alpha_{3,j}} 10 \Bigg(\binom{c_{k,3,j} + d_{k,3,j}}{2} \\
&\qquad \Big(p_j + \rho_{k-1,3,j}[1] + \rho_{k-1,3,j}[2]\Big)^2\Bigg)\Bigg)\Bigg), \\
&\overset{(2)}{\geq} \quad \frac{\mathbf{P}(2^n, \gamma_{3,j}) \cdot \mathbf{P}(2^n - p_j, \delta_{3,j})}{2^{n\sum\limits_{i\in\mathcal{I}_3^{=}[j]} q_i}} \Bigg(1 - \frac{1}{2^{2n}} \cdot \Bigg(\sum_{k=1}^{\alpha_{3,j}} 10 \Bigg(\binom{c_{k,3,j} + d_{k,3,j}}{2} \\
&\qquad \Big(p_j + 4q_{3,j}\Big)^2\Bigg)\Bigg)\Bigg),
\end{aligned}$$

where (1) and (2) follows due to the fact that $\rho_{k-1,b,j}[1] \leq 2q_{b,j}$ and $\rho_{k-1,b,j}[2] \leq 2q_{b,j}$ for $b \in \{2,3\}$. Moreover, we have

$$q_{2,j} := \sum_{i\in\mathcal{I}_2^{=}[j]} q_i,\ q_{3,j} := \sum_{i\in\mathcal{I}_3^{=}[j]} q_i.$$

For a fixed $j \in [r']$, let β_j denotes the number of components of the graph $\cup_{i\in\mathcal{I}^{\neq}[j]} H_i$. Let $c'_{k,j}$ (resp. $d'_{k,j}$) denotes the number of vertices corresponding to the Σ (resp. Θ) values in the k-th component of the graph $\cup_{i\in\mathcal{I}^{\neq}[j]} H_i$, where $k \in [\beta_j]$. Therefore, we have

$$\eta_j = \sum_{k=1}^{\beta_j} c'_{k,j},\ \xi_j = \sum_{k=1}^{\beta_j} d'_{k,j}.$$

Moreover, following the notations of Sect. 2.4, we write

$$\rho'_{m,j}[1] = \sum_{k=1}^{m} c'_{k,j},\ \rho'_{m,j}[2] = \sum_{k=1}^{m} d'_{k,j},$$

where $m \in [\beta_j]$. Finally, for a fixed j, we have

$$\begin{aligned}
h(\cup_{i\in\mathcal{I}^{\neq}[j]} H_i) \quad &\geq \quad \frac{\mathbf{P}(2^n,\eta_j)\cdot\mathbf{P}(2^n,\xi_j)}{2^{n\sum_{i\in\mathcal{I}^{\neq}[j]} q_i}}\Bigg(1-\frac{1}{2^{2n}}\cdot\Bigg(\sum_{k=1}^{\beta_j} 10\Bigg(\binom{c'_{k,j}+d'_{k,j}}{2} \\
&\qquad \Big(\rho'_{k-1,j}[1]+\rho'_{k-1,j}[2]\Big)^2\Bigg)\Bigg)\Bigg) \\
&\overset{(3)}{\geq} \quad \frac{\mathbf{P}(2^n,\eta_j)\cdot\mathbf{P}(2^n,\xi_j)}{2^{n\sum_{i\in\mathcal{I}^{\neq}[j]} q_i}}\Bigg(1-\frac{1}{2^{2n}}\cdot\Bigg(\sum_{k=1}^{\beta_j} 160 q_j'^2\binom{c'_{k,j}+d'_{k,j}}{2}\Bigg)\Bigg),
\end{aligned}$$

where (3) follows due to the fact that $\rho'_{k-1,j}[1] \leq 2q_j$ and $\rho'_{k-1,j}[2] \leq 2q_j$. Moreover, we have

$$q'_j := \sum_{i\in\mathcal{I}^{\neq}[j]} q_i.$$

By plugging in the inequalities (1), (2), and (3) into Eqn. (36) and then by taking the ratio of real to ideal interpolation probability, we obtain

$$\frac{\Pr[\mathsf{X}_{\mathrm{re}}=\tau]}{\Pr[\mathsf{X}_{\mathrm{id}}=\tau]} \geq \prod_{i=1}^{u} 2^{nq_i}\cdot \tag{37}$$

$$\begin{aligned}
&\prod_{j=1}^{r_2}\frac{1}{2^{n\sum_{i\in\mathcal{I}_2^{=}[j]} q_i}}\Bigg(1-\frac{1}{2^{2n}}\cdot\Bigg(\sum_{k=1}^{\alpha_{2,j}} 10\Bigg(\binom{c_{k,2,j}+d_{k,2,j}}{2}\Big(p_j+4q_{2,j}\Big)^2\Bigg)\Bigg)\Bigg)\cdot \\
&\prod_{j=1}^{r_3}\frac{1}{2^{n\sum_{i\in\mathcal{I}_3^{=}[j]} q_i}}\Bigg(1-\frac{1}{2^{2n}}\cdot\Bigg(\sum_{k=1}^{\alpha_{3,j}} 10\Bigg(\binom{c_{k,3,j}+d_{k,3,j}}{2}+\Big(p_j+4q_{3,j}\Big)^2\Bigg)\Bigg)\Bigg)\cdot \\
&\prod_{j\in r'}\frac{1}{2^{n\sum_{i\in\mathcal{I}^{\neq}[j]} q_i}}\Bigg(1-\frac{1}{2^{2n}}\cdot\Bigg(\sum_{k=1}^{\beta_j} 160 q_j'^2\Bigg(\binom{c'_{k,j}+d'_{k,j}}{2}\Bigg)\Bigg)\Bigg) \\
&\geq \Bigg(\prod_{j=1}^{r_2}\frac{2^{n\sum_{i\in\mathcal{I}_2^{=}[j]} q_i}}{2^{n\sum_{i\in\mathcal{I}_2^{=}[j]} q_i}}\Bigg)\cdot\Bigg(\prod_{j=1}^{r_3}\frac{2^{n\sum_{i\in\mathcal{I}_3^{=}[j]} q_i}}{2^{n\sum_{i\in\mathcal{I}_3^{=}[j]} q_i}}\Bigg)\cdot\Bigg(\prod_{j\in r'}\frac{2^{n\sum_{i\in\mathcal{I}^{\neq}[j]} q_i}}{2^{n\sum_{i\in\mathcal{I}^{\neq}[j]} q_i}}\Bigg) \\
&\cdot\Bigg(1-\sum_{j=1}^{r_2}\frac{1}{2^{2n}}\cdot\Bigg(\sum_{k=1}^{\alpha_{2,j}} 10\Bigg(\binom{c_{k,2,j}+d_{k,2,j}}{2}\Big(p_j+4q_{2,j}\Big)^2\Bigg)\Bigg)\Bigg) \\
&\cdot\Bigg(1-\sum_{j=1}^{r_3}\frac{1}{2^{2n}}\cdot\Bigg(\sum_{k=1}^{\alpha_{3,j}} 10\Bigg(\binom{c_{k,3,j}+d_{k,3,j}}{2}\Big(p_j+4q_{3,j}\Big)^2\Bigg)\Bigg)\Bigg) \\
&\cdot\Bigg(1-\sum_{j=1}^{r'}\frac{1}{2^{2n}}\cdot\Bigg(\sum_{k=1}^{\beta_j} 160 q_j'^2\binom{c'_{k,j}+d'_{k,j}}{2}\Bigg)\Bigg).
\end{aligned} \tag{38}$$

Let,

$$\begin{aligned}
\Phi_{2,j}(\tau) \quad &:= \quad \frac{1}{2^{2n}}\cdot\Bigg(\sum_{k=1}^{\alpha_{2,j}} 10\Bigg(\binom{c_{k,2,j}+d_{k,2,j}}{2}\Big(p_j+4q_{2,j}\Big)^2\Bigg)\Bigg) \\
\Phi_{3,j}(\tau) \quad &:= \quad \frac{1}{2^{2n}}\cdot\Bigg(\sum_{k=1}^{\alpha_{3,j}} 10\Bigg(\binom{c_{k,3,j}+d_{k,3,j}}{2}\Big(p_j+4q_{3,j}\Big)^2\Bigg)\Bigg)
\end{aligned}$$

$$\Phi'_j(\tau) \quad := \quad \frac{1}{2^{2n}} \cdot \left(\sum_{k=1}^{\beta_j} 160 q_j'^2 \binom{c'_{k,j} + d'_{k,j}}{2} \right) \Bigg)$$

Therefore, we have

$$\frac{\Pr[\mathsf{X}_{\mathrm{re}} = \tau]}{\Pr[\mathsf{X}_{\mathrm{id}} = \tau]} \quad \geq \quad 1 - \left(\sum_{j=1}^{r_2} \Phi_{2,j}(\tau) + \sum_{j=1}^{r_3} \Phi_{3,j}(\tau) + \sum_{j=1}^{r'} \Phi'_j(\tau) \right). \tag{39}$$

<u>Computing Expectation.</u>

$$\mathbf{E}[\Phi_{2,j}(\mathsf{X}_{\mathrm{id}})] \overset{(4)}{=} \frac{1}{2^{2n}} \cdot \left(10(p_j + 4q_{2,j})^2 \mathbf{E}\left[\sum_{k=1}^{\alpha_{2,j}} \binom{c_{k,2,j} + d_{k,2,j}}{2} \right] \right)$$

$$\mathbf{E}[\Phi_{3,j}(\mathsf{X}_{\mathrm{id}})] \overset{(5)}{=} \frac{1}{2^{2n}} \cdot \left(10(p_j + 4q_{3,j})^2 \mathbf{E}\left[\sum_{k=1}^{\alpha_{3,j}} \binom{c_{k,3,j} + d_{k,3,j}}{2} \right] \right)$$

$$\mathbf{E}[\Phi'_j(\mathsf{X}_{\mathrm{id}})] \overset{(6)}{=} \frac{1}{2^{2n}} \cdot \left(160 q_j'^2 \left(\mathbf{E}\left[\sum_{k=1}^{\beta_j} \binom{c'_{k,j} + d'_{k,j}}{2} \right] \right) \right)$$

We first compute the right-hand side of (4). Let $\tilde{c}_{k,2,j} := c_{k,2,j} - 1$ and $\tilde{d}_{k,2,j} := d_{k,2,j} - 1$. Therefore, we have

$$\begin{aligned} \mathbf{E}\left[\sum_{k=1}^{\alpha_{2,j}} \binom{c_{k,2,j} + d_{k,2,j}}{2} \right] &= \mathbf{E}\left[\sum_{k=1}^{\alpha_{2,j}} \binom{\tilde{c}_{k,2,j} + \tilde{d}_{k,2,j}}{2} \right] + 2\mathbf{E}\left[\sum_{k=1}^{\alpha_{2,j}} (\tilde{c}_{k,2,j} + \tilde{d}_{k,2,j}) \right] \\ &\overset{(7)}{\leq} \mathbf{E}\left[\sum_{k=1}^{\alpha_{2,j}} \binom{\tilde{c}_{k,2,j} + \tilde{d}_{k,2,j}}{2} \right] + 8q_{2,j}. \end{aligned} \tag{40}$$

where the inequality (7) holds as $\tilde{c}_{k,2,j} \leq 2q_{2,j}$ and $\tilde{d}_{k,2,j} \leq 2q_{2,j}$. Let us define an indicator random variable $\mathbb{I}_{k,2,j}[ab]$ which takes the value 1 if $\Sigma_a^{i_1} = \Sigma_b^{i_2}$ or $\Theta_a^{i_1} = \Theta_b^{i_2}$, where $i_1, i_2 \in \mathcal{I}_2^{=}[j]$. Therefore, we have

$$\begin{aligned} \mathbf{E}\left[\sum_{k=1}^{\alpha_{2,j}} \binom{\tilde{c}_{k,2,j} + \tilde{d}_{k,2,j}}{2} \right] &= \sum_{i \in \mathcal{I}_2^{=}[j]} \sum_{a \neq b}^{q_i} \mathbf{E}\Big[\mathbb{I}_{k,2,j}[ab] = 1 \Big] \\ &= \sum_{i \in \mathcal{I}_2^{=}[j]} \sum_{a \neq b}^{q_i} (\Pr[\Sigma_a^i = \Sigma_b^i] + \Pr[\Theta_a^i = \Theta_b^i]) \\ &\overset{(8)}{\leq} \sum_{i \in \mathcal{I}_2^{=}[j]} 2q_i^2/2^n \leq 2q_{2,j}^2/2^n, \end{aligned} \tag{41}$$

where (8) follows from Lemma 4 and Lemma 8. By plugging-in the bound of Eqn. (41) into Eqn. (40), we have

$$\mathbf{E}\left[\sum_{k=1}^{\alpha_{2,j}} \binom{c_{k,2,j} + d_{k,2,j}}{2} \right] \quad \leq \quad 2q_{2,j}^2/2^n + 8q_{2,j}, \tag{42}$$

In a similar way, we have

$$\mathbf{E}\left[\sum_{k=1}^{\alpha_{3,j}} \binom{c_{k,3,j} + d_{k,3,j}}{2} \right] \quad \leq \quad 2q_{3,j}^2/2^n + 8q_{3,j} \tag{43}$$

$$\mathbf{E}\Big[\sum_{k=1}^{\beta_j}\binom{c'_{k,j}+d'_{k,j}}{2}\Big] \quad\leq\quad 2q_j'^2/2^n + 8q'_j. \tag{44}$$

By plugging-in the bound of Eqn. (42), Eqn. (43), and Eqn. (44) into the right-hand side of Eqn. (4), Eqn. (5), and Eqn. (6) respectively, we have

$$\mathbf{E}[\Phi_{2,j}(\mathsf{X}_{\mathrm{id}})] \quad\leq\quad \frac{1}{2^{2n}}\cdot\Big(10(p_j+4q_{2,j})^2(2q_{2,j}^2/2^n+8q_{2,j})\Big) \tag{45}$$

$$\mathbf{E}[\Phi_{3,j}(\mathsf{X}_{\mathrm{id}})] \quad\leq\quad \frac{1}{2^{2n}}\cdot\Big(10(p_j+4q_{3,j})^2(2q_{3,j}^2/2^n+8q_{3,j})\Big) \tag{46}$$

$$\mathbf{E}[\Phi'_j(\mathsf{X}_{\mathrm{id}})] \quad\leq\quad 320q_j'^4/2^{3n}+1280q_j'^3/2^{2n}. \tag{47}$$

By doing a simple algebra on Eqn. (45)-Eqn. (47) and by using Lemma 9 and equality

$$\sum_{j\in[r_2]} q_{2,j} + \sum_{j\in[r_3]} q_{3,j} + \sum_{j\in[r']} q'_j = q \text{ and } \sum_{j=1}^{g} p_j = p,$$

we derive the result.

6.3 Proof of Theorem 6

In the proof of Theorem 5, the only term that carries an ℓ factor, is $q^{3/2}\ell k/2^n$, which appears while bounding the probability of the joint event E and $|\mathcal{S}|<\mu$ under the bad events $\mathsf{Bad3}$ and $\mathsf{Bad4}$ in Eqn. (65). To obtain an ℓ-free bound, we bound the bad events $\mathsf{Bad3}$ and $\mathsf{Bad4}$ in a different way. [4]

Bounding Bad3. We bound the event $\mathsf{Bad3}$ in the same way as we did for proving the security of LightMAC_Plus. However, in this case, when $k\geq 4n/3$, we choose the value of $\mu = 2^{3k/4-n}k\ell$. With this value of μ, we have

$$\begin{aligned}
\Pr[\mathsf{Bad3}] \quad&\leq\quad \frac{p\sqrt{q}}{2^k}+\sum_{i=\sqrt{q}}^{u}\frac{q_i^2\mu}{2^k}+\Pr[|\mathcal{S}|\geq\mu] \overset{(1)}{\leq} \frac{p\sqrt{q}}{2^k}+\frac{q^{3/2}\mu}{2^k}+2^{2n}\cdot p^{2\ell}\cdot\Big(\frac{6p}{2^n\mu}\Big)^{\mu}\\
&\overset{(2)}{\leq}\quad \frac{p\sqrt{q}}{2^k}+\frac{q^{3/2}}{2^n}\cdot\frac{\ell k}{2^{k/4}}+2^{2n}\cdot p^{2\ell}\cdot\Big(\frac{6p}{2^{3k/4}k\ell}\Big)^{2^{3k/4-n}k\ell}\\
&\overset{(3)}{\leq}\quad \frac{p\sqrt{q}}{2^k}+\frac{q^{3/2}}{2^n}+\frac{6p}{2^{3k/4}},
\end{aligned} \tag{48}$$

where inequality (1) follows as the summation q_i^2, for $i\in[\sqrt{q},u]$ is bounded above by $q^{3/2}$ due to Lemma 2 and we inheriting the bound of the probability of the event $|\mathcal{S}|\geq\mu$ from the previous analysis. Moreover, inequality (2) follows by choosing the value of $\mu = 2^{3k/4-n}k\ell$.

Using a similar argument, we bound the probability of event $\mathsf{Bad4}$ as follows:

$$\Pr[\mathsf{Bad4}] \quad\leq\quad \frac{p\sqrt{q}}{2^k}+\frac{q^{3/2}}{2^n}+\frac{6p}{2^{3k/4}}. \tag{49}$$

By inheriting the bounds of the remaining bad events from the previous analysis and simplifying, we obtain the probability of the bad transcript as follows:

$$\Pr[\mathsf{X}_{\mathrm{id}}\in\mathsf{BadT}] \quad\leq\quad \frac{2u}{2^k}+\frac{u^3}{6.2^{2k}}+\frac{2up^2}{2^{2k}}+\frac{6q^2\ell}{2^{n+k}}+\frac{4qq_{\max}^2}{2^{2n}}+\frac{2p\sqrt{q}}{2^k}$$

[4] We would like to note that the term $pq\ell k/2^{n+k}$ also carries an ℓ factor which arises while bounding sub case $\mathsf{E.1}$ under the bad event $\mathsf{Bad1}$ and $\mathsf{Bad2}$. However, this term does not create any problem in achieving an ℓ-free bound if we appropriately set the bound on ℓ. Unfortunately, this is not the case for the term $q^{3/2}\ell k/2^n$, and thus, we need a separate treatment with this bound to achieve ℓ-freeness.

$$+\frac{2q^{3/2}}{2^n} + \frac{2q\sqrt{q_{\max}}}{2^n} + \min\left\{\frac{2up}{2^k}, \frac{2q}{2^{2n/3}}\right\} + \frac{13p}{2^{3k/4}}, \qquad (50)$$

assuming $k \geq 4n/3$ and $\ell \leq 2^{n/3}/k$. Finally, by combining Eqn. (50) and following the analysis of lower bounding the ratio of real to ideal interpolation probability of a good transcript, we obtain the desired ℓ-free security bound of the LightMAC_Plus construction in the multi-user setting.

7 Conclusion

In this paper we have analyzed multi-user security of LightMAC and LightMAC_Plus construction. We have shown that likewise, the security of LightMAC construction in the single-user model achieves birthday-bound security in the multi-user security model. However, we have been able to show that LightMAC_Plus achieves $2n/3$-bit security in the multi-user setup, whereas it has $3n/4$-bit security in the single-user setting, and the bound is tight. Thus, it remains open to improving the multi-user security bound of LightMAC_Plus from $2n/3$ bits to $3n/4$ bits.

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Appendix

Proof of Lemma 3

Consider the first component C_1 of the graph G. Let $Y_{i_1} \in \mathcal{V}_1$ be any arbitrary vertex of C_1. There are $2^n - \Delta_1$ choices for assigning values to the variable Y_{i_1}. Let the assigned value to Y_{i_1} be y_{i_1}. Now, for any other variable Y_{i_2} of C_1, we consider the path $\mathcal{P}$ from Y_{i_1} to Y_{i_2} and assign the value $y_{i_1} \oplus \mathcal{L}(\mathcal{P})$ to the variable Y_{i_2}. Let this value be y_{i_2}. Note that the path is unique as the graph is acyclic and $\mathcal{L}(\mathcal{P}) \neq 0^n$ and hence $y_{i_1} \neq y_{i_2}$. However, we require $y_{i_2} \notin \mathcal{S}_1$. Similarly, for any other variable Z_{i_2} of C_1, we consider the path $\mathcal{P}$ from Y_{i_1} to Z_{i_2} and assign the value $y_{i_1} \oplus \mathcal{L}(\mathcal{P})$ to the variable Z_{i_2}. Let this value be z_{i_2}. Note that $z_{i_1} \neq z_{i_2}$. However, we require $z_{i_2} \notin \mathcal{S}_2$. Thus, the number of valid choices for assigning values to Y_{i_1} is at least $2^n - (c_1\Delta_1 + d_1\Delta_2)$. In general, for the i-th component, the number of ways we assign a value to a vertex in $\mathcal{V}_1$ of component C_i is at least

$$2^n - \big((c_1 + \ldots + c_{i-1} + \Delta_1)c_i + (d_1 + \ldots + d_{i-1} + \Delta_2)d_i\big).$$

This is because, to assign a value in a vertex in $\mathcal{V}_1$ of C_i, we need to ensure that the assignment should not create any collision between the assigned values and all the previously assigned values in all vertices of $\mathcal{V}_1$ of component $\mathsf{C}_1, \ldots, \mathsf{C}_{i-1}$. Similarly, the assigned values should not create any collision between the assigned values in the vertices $\mathcal{V}_2$ of C_i and all the previously assigned values in all vertices of $\mathcal{V}_2$ of component $\mathsf{C}_1, \ldots, \mathsf{C}_{i-1}$. Additionally, the assigned vertex should not collide with any element of Δ_1 or Δ_2. Let $h(\alpha)$ denotes the number of solutions chosen outside of $\mathcal{S}_1$ and $\mathcal{S}_2$ to $\mathcal{E}_{\mathsf{G}}$. Therefore, we have

$$\begin{aligned} h(\alpha) &\geq \prod_{i=1}^{\alpha} (2^n - ((c_1 + \ldots + c_{i-1} + \Delta_1)c_i + (d_1 + \ldots + d_{i-1} + \Delta_2)d_i)) \\ &= \prod_{i=1}^{\alpha} (2^n - ((\rho_{i-1}[1] + \Delta_1)c_i + (\rho_{i-1}[2] + \Delta_2)d_i)) \,. \end{aligned} \tag{51}$$

We would like to note that since the graph G is good, for each component i, at least one of c_i or d_i must be 1. By multiplying $2^{nq}/\mathbf{P}(2^n - \Delta_1, s_\ell)\mathbf{P}(2^n - \Delta_2, s_{\mathtt{r}})$ in both side of Eqn. (51), we have

$$\begin{aligned} h(\alpha) \quad &\cdot \quad \frac{2^{nq}}{\mathbf{P}\left(2^n - \Delta_1, s_\ell\right) \mathbf{P}\left(2^n - \Delta_2, s_{\mathtt{r}}\right)} \\ &\geq \prod_{i=1}^{\alpha} \frac{\left(2^n - \left(\left(\rho_{i-1}[1] + \Delta_1\right) c_i + \left(\rho_{i-1}[2] + \Delta_2\right) d_i\right)\right) 2^{n(c_i+d_i-1)}}{\mathbf{P}\left(2^n - \left(\Delta_1 + \rho_{i-1}[1]\right), c_i\right) \mathbf{P}\left(2^n - \left(\Delta_2 + \rho_{i-1}[2]\right), d_i\right)} \\ &= \prod_{i=1}^{\alpha} \frac{2^{n(c_i+d_i)} - 2^{n(c_i+d_i-1)}\left(\left(\rho_{i-1}[1] + \Delta_1\right) c_i + \left(\rho_{i-1}[2] + \Delta_2\right) d_i\right)}{\mathbf{P}\left(2^n - \left(\Delta_1 + \rho_{i-1}[1]\right), c_i\right) \mathbf{P}\left(2^n - \left(\Delta_2 + \rho_{i-1}[2]\right), d_i\right)}. \end{aligned}$$

Now, we note that

$$\begin{aligned} \underbrace{\mathbf{P}\left(2^n - \left(\Delta_1 + \rho_{i-1}[1]\right), c_i\right)}_{A_i} \leq 2^{nc_i} &- 2^{n(c_i-1)} \left(\left(\Delta_1 + \rho_{i-1}[1]\right) c_i + \binom{c_i}{2}\right) \\ &+ 2^{n(c_i-2)} \left(\binom{c_i}{2} (\Delta_1 + \rho_{i-1}[1])^2 \right. \\ &\left. + \binom{c_i}{2} (c_i - 1)(\Delta_1 + \rho_{i-1}[1]) \binom{c_i}{2} \frac{(c_i - 2)(3c_i - 1)}{12} \right). \end{aligned} \tag{52}$$

Similarly, we have

$$\underbrace{\mathbf{P}(2^n-(\Delta_2+\rho_{i-1}[2]),d_i)}_{B_i} \le 2^{nd_i}-2^{n(d_i-1)}\left((\Delta_2+\rho_{i-1}[2])d_i+\binom{d_i}{2}\right)$$
$$+2^{n(d_i-2)}\left(\binom{d_i}{2}(\Delta_2+\rho_{i-1}[2])^2\right.$$
$$\left.+\binom{d_i}{2}(d_i-1)(\Delta_2+\rho_{i-1}[2])+\binom{d_i}{2}\frac{(d_i-2)(3d_i-1)}{12}\right). \tag{53}$$

Let us define the following:

$$C_i := \left(\binom{c_i}{2}(\Delta_1+\rho_{i-1}[1])^2+\binom{c_i}{2}(c_i-1)(\Delta_1+\rho_{i-1}[1])+\right.$$
$$\left.\binom{c_i}{2}\frac{(c_i-2)(3c_i-1)}{12}\right),$$
$$D_i := \left(\binom{d_i}{2}(\Delta_2+\rho_{i-1}[2])^2+\binom{d_i}{2}(d_i-1)(\Delta_2+\rho_{i-1}[2])+\right.$$
$$\left.\binom{d_i}{2}\frac{(d_i-2)(3d_i-1)}{12}\right).$$

Combining Eqn. (52) and Eqn. (53), and by assuming $\big(\Delta_1+\rho_\alpha[1]\big)c_{\max}\le 2^{n-3}$, we have

$$A_iB_i \le 2^{n(c_i+d_i)}-2^{n(c_i+d_i-1)}\underbrace{\left(c_i(\Delta_1+\rho_{i-1}[1])+d_i(\Delta_2+\rho_{i-1}[2])+\binom{c_i}{2}+\binom{d_i}{2}\right)}_{Y_i}$$
$$+2^{n(c_i+d_i-2)}\underbrace{\left(C_i+D_i+\left(c_i(\Delta_1+\rho_{i-1}[1])+\binom{c_i}{2}\right)\left(d_i(\Delta_2+\rho_{i-1}[2])+\binom{d_i}{2}\right)\right)}_{X_i}, \tag{54}$$

where we use the fact that, $2^n[D_i\big(c_i(\Delta_1+\rho_{i-1}[1])+\binom{c_i}{2}\big)+C_i\big(d_i(\Delta_2+\rho_{i-1}[2])+\binom{d_i}{2}\big)]-C_iD_i\ge 0$. By using Eqn. (54), we have

$$h(\alpha)\cdot\frac{2^{nq}}{\mathbf{P}(2^n-\Delta_1,s_\ell)\mathbf{P}(2^n-\Delta_2,s_\mathbf{r})}$$
$$\ge\prod_{i=1}^{\alpha}1+\frac{2^{n(c_i+d_i-1)}\left(\binom{c_i}{2}+\binom{d_i}{2}\right)-2^{n(c_i+d_i-2)}X_i}{2^{n(c_i+d_i)}-2^{n(c_i+d_i-1)}Y_i+2^{n(c_i+d_i-2)}X_i}$$
$$=\prod_{i=1}^{\alpha}1+\frac{2^n\left(\binom{c_i}{2}+\binom{d_i}{2}\right)-X_i}{2^{2n}-2^nY_i+X_i}\overset{(2)}{\ge}\prod_{i=1}^{\alpha}\left(1-\frac{2X_i}{2^{2n}}\right).$$

Thus, we have

$$h(\alpha)\overset{(3)}{\ge}\frac{\mathbf{P}(2^n-\Delta_1,s_\ell)\cdot\mathbf{P}(2^n-\Delta_2,s_\mathbf{r})}{2^{nq}}\times$$
$$\left(1-\sum_{i=1}^{\alpha}\frac{10\binom{c_i+d_i}{2}\left(\Delta_1+\Delta_2+\rho_{i-1}[1]+\rho_{i-1}[2]\right)^2}{2^{2n}}\right)$$

where (2) follows from the fact that, $2^nY_i-X_i\le 2^{2n-1}$ which holds true by assuming $(\Delta_1+\rho_\alpha[1])c_{\max}\le 2^{n-3}$ and $(\Delta_2+\rho_\alpha[2])d_{\max}\le 2^{n-3}$. Moreover, inequality (3) holds as it can be easily seen that each term of X_i is at most $\binom{c_i+d_i}{2}(\Delta_1+\Delta_2+\rho_{i-1}[1]+\rho_{i-1}[2])^2$.

A Proof of Lemma 9

Proof. Let us define the event

$$\begin{aligned}\mathsf{BadT} \quad := \quad & \underbrace{\left(\vee_{i=1}^{8}\mathsf{BadKi}\right) \vee \left(\vee_{i=1}^{4}\ \mathsf{Badi}\right)}_{\mathsf{BadK}} \vee \left(\mathsf{Bad5} \mid \overline{\mathsf{Bad3}}\right) \vee \left(\mathsf{Bad6} \mid \overline{\mathsf{Bad4}}\right) \\ & \vee \left(\ \vee_{i=7}^{10}\ \mathsf{Badi} \mid \overline{\mathsf{BadK4}}\right) \vee \left(\ \vee_{i=11}^{12}\ (\mathsf{Badi} \mid \overline{\mathsf{Bad3}} \wedge \overline{\mathsf{Bad4}})\right) \\ & \vee \left(\mathsf{Bad13} \vee \mathsf{Bad14}\right).\end{aligned}$$

We upper-bound the probability of individual bad events in the ideal world, and then, using the union bound, we sum up the bounds to obtain the overall bound on the probability of bad transcripts in the ideal world.

I. Bounding $\mathsf{BadK1}, \mathsf{BadK2}$ and $\mathsf{BadK3}$. Recall that the event $\mathsf{BadK1}$ holds if there exists two distinct users i_1 and i_2 such that $K_1^{i_1}$ collides with $K_1^{i_2}$ and $K_2^{i_1}$ collides with $K_2^{i_2}$. In the ideal world, since the block cipher keys of each user are drawn independently and uniformly at random, for a fixed choice of two users i_1 and i_2, the probability that $K_1^{i_1} = K_1^{i_2}, K_2^{i_1} = K_2^{i_2}$ holds, is exactly 2^{-2k}. Therefore, by varying over all possible choices of users, we have

$$\Pr[\mathsf{BadK1}] \leq \frac{\binom{u}{2}}{2^{2k}} \leq \frac{u^2}{2^{2k+1}}. \tag{55}$$

Since the events $\mathsf{BadK2}$ and $\mathsf{BadK3}$ are exactly similar to the event $\mathsf{BadK1}$, we bound these two events similarly, and hence, we have

$$\Pr[\mathsf{BadK2}] \leq \frac{u^2}{2^{2k+1}},\ \Pr[\mathsf{BadK3}] \leq \frac{u^3}{6.2^{2k}}. \tag{56}$$

II. Bounding $\mathsf{BadK4}$. Recall that the event $\mathsf{BadK4}$ holds, if there exists two distinct users i_1 and i_2 such that $K_b^{i_1}$ collides with $K_b^{i_2}$, $K_1^{i_1}$ collides with J^j, and $K_1^{i_2}$ collides with $J^{j'}$ for some $j, j' \in [g]$ and $b \in \{2,3\}$. In the ideal world, since the block cipher keys of each user are drawn independently and uniformly at random, for a fixed choice of two users i_1 and i_2, and for a fixed $b \in \{2,3\}$, the probability that $K_b^{i_1} = K_b^{i_2}, K_1^{i_1} = J^j, K_1^{i_2} = J^{j'}$ holds, is exactly 2^{-3k} by using the randomness of $K_1^{i_1}, K_1^{i_2}$, and $K_b^{i_1}$. Therefore, by varying over all possible choices of users and $b \in \{2,3\}$, we have

$$\Pr[\mathsf{BadK4}] \leq \frac{2\binom{u}{2}g^2}{2^{3k}} \leq \frac{2\binom{u}{2}p^2}{2^{3k}} \leq \frac{u^2p^2}{2^{3k}}. \tag{57}$$

III. Bounding $\mathsf{BadK5}$. Recall that the event $\mathsf{BadK5}$ holds, if there exists an user i such that K_2^i collides with a chosen ideal-cipher key J^j and K_3^i collides with an another chosen ideal-cipher key $J^{j'}$, for some $j, j' \in [g]$. In the ideal world, since the block cipher keys of each user are drawn independently and uniformly at random, for a fixed choice of user i and $j, j' \in [g]$, the probability that $K_2^i = J^j, K_3^i = J^{j'}$ holds, is exactly 2^{-2k} by using the randomness of K_2^i and K_3^i. Therefore, by varying over all possible choices of users $j, j' \in [g]$, we have

$$\Pr[\mathsf{BadK5}] \leq \frac{ug^2}{2^{2k}} \leq \frac{up^2}{2^{2k}}. \tag{58}$$

IV. Bounding $\mathsf{BadK6}$. Recall that the event $\mathsf{BadK6}$ holds, if there exists an user i such that K_2^i collides with K_3^i. In the ideal world, since the block cipher keys of each user are drawn independently and uniformly at random, for a fixed choice of user i, the probability that $K_2^i = K_3^i$ holds is exactly 2^{-k} using the randomness of K_2^i. Therefore, by varying over all possible choices of users, we have,

$$\Pr[\mathsf{BadK6}] \leq \frac{u}{2^k}. \tag{59}$$

V. Bounding BadK7. Recall that the event $\mathsf{BadK7}$ holds, if there exists two distinct users i_1, i_2 and two construction queries of the users i_1 and i_2 such that $K_1^{i_1}$ collides with $K_2^{i_2}$ and $\langle m\rangle \| M_a^{i_1}[m] = \Sigma_b^{i_2}$. In the ideal world, the block cipher keys of each user are drawn independently and uniformly at random. Therefore, for a fixed choice of a query of users (i_1, a) and (i_2, b) and a fixed choice of $m \in [\ell_a^{i_1}]$ (given that ℓ is the maximum number of message blocks and so $\ell^{i_1} \leq \ell$), the probability that $K_1^{i_1} = K_2^{i_2}, \langle m\rangle \| M_a^{i_1}[m] = \Sigma_b^{i_2}$ holds, is exactly $2^{-(n+k)}$, where the bound on the probability of the event $\langle m\rangle \| M_a^{i_1}[m] = \Sigma_b^{i_2}$ follows from Lemma 4. Therefore, by varying over all possible choices of indices, we have

$$\Pr[\mathsf{BadK7}] \leq \sum_{i_2=1}^{u} \sum_{b=1}^{q_{i_2}} \sum_{i_1=1}^{u} \sum_{a=1}^{q_{i_1}} \sum_{m=1}^{\ell_a^{i_1}} \frac{1}{2^{n+k}} \leq \frac{q^2 \ell}{2^{n+k}}. \tag{60}$$

VI. Bounding BadK8. The event $\mathsf{BadK8}$ is similar to the event $\mathsf{BadK7}$. So, we similarly bound the event, and we have

$$\Pr[\mathsf{BadK8}] \leq \frac{q^2 \ell}{2^{n+k}}. \tag{61}$$

VII. Bounding Bad1 and Bad2. Recall that the event $\mathsf{Bad1}$ and $\mathsf{Bad2}$ is identical to the event $\mathsf{Bad2}$ of the $\mathsf{LightMAC}$ construction. Therefore, we bound these two events in the same way, and hence, following the analysis of the $\mathsf{Bad2}$ event of the $\mathsf{LightMAC}$ construction, we have

$$\Pr[\mathsf{Bad1}] \leq \min\left\{\frac{up}{2^k}, \frac{q\ell k}{2^n}\right\} + \frac{14p}{2^k}, \tag{62}$$

$$\Pr[\mathsf{Bad2}] \leq \min\left\{\frac{up}{2^k}, \frac{q\ell k}{2^n}\right\} + \frac{14p}{2^k}, \tag{63}$$

VIII. Bounding Bad3 and Bad4. To bound the event $\mathsf{Bad3}$, we consider two cases separately: (a) when $i < \sqrt{q}$ and (b) when $i \geq \sqrt{q}$. For the first case, we can easily bound the event by just considering the sub-event $K_1^i = J^j$, which holds with probability $\sqrt{q}p/2^k$. On the other hand, when $i \geq \sqrt{q}$, then we need to consider both $\Sigma_a^i = \Sigma_b^i$ and $K_1^i = J^j$.

$$\begin{aligned}
\Pr[\mathsf{Bad3}] &\leq \sum_{i=1}^{\sqrt{q}-1} \Pr[\exists j \in [g] : K_1^i = J^j] + \sum_{i=\sqrt{q}}^{u} \sum_{a,b=1}^{q_i} \Pr[\exists j \in [g] : \Sigma_a^i = \Sigma_b^i, K_1^i = J^j] \\
&\leq \frac{p\sqrt{q}}{2^k} + \sum_{i=\sqrt{q}}^{u} \sum_{a,b=1}^{q_i} \Pr[\underbrace{\exists j \in [g] : \Sigma_a^i = \Sigma_b^i, K_1^i = J^j}_{\mathsf{E}}].
\end{aligned} \tag{64}$$

To bound the probability of the latter event E, we define the set

$$\mathcal{S} \triangleq \{j \in [g] : \Sigma_{J^j}(M) = \Sigma_{J^j}(M')\}.$$

Therefore, we can write

$$\sum_{i=\sqrt{q}}^{u} \sum_{a,b=1}^{q_i} \Pr[\mathsf{E}] = \sum_{i=\sqrt{q}}^{u} \sum_{a,b=1}^{q_i} \Pr[\mathsf{E}, |\mathcal{S}| < \mu] + \Pr[|\mathcal{S}| \geq \mu]. \tag{65}$$

By combining Eqn. (64) and Eqn. (65), we have

$$\begin{aligned}\Pr[\mathsf{Bad3}] &\leq \frac{p\sqrt{q}}{2^k} + \sum_{i=\sqrt{q}}^{u} \frac{q_i^2 \mu}{2^k} + \Pr[|\mathcal{S}| \geq \mu] \overset{(1)}{\leq} \frac{p\sqrt{q}}{2^k} + \frac{q^{3/2}\mu}{2^k} + 2\left(\frac{6p}{2^k}\right)^{2^{k-n}k\ell} \\ &\overset{(2)}{\leq} \frac{p\sqrt{q}}{2^k} + \frac{q^{3/2}k\ell}{2^n} + 2\left(\frac{6p}{2^k}\right)^{2^{k-n}k\ell}, \end{aligned} \tag{66}$$

where inequality (1) follows as the summation q_i^2, for $i \in [\sqrt{q}, u]$ is bounded above by $q^{3/2}$ due to Lemma 2 and we inherit the bound of the probability of the event $|\mathcal{S}| \geq \mu$ from the previous analysis. Moreover, inequality (2) follows by choosing the value of $\mu = 2^{k-n}k\ell$.

Bounding the event Bad4, is exactly identical to that of Bad3 and hence, we bound the event Bad4 in exactly the same way as we did for Bad3, and hence, following the analysis of the Bad3, we have

$$\Pr[\mathsf{Bad4}] \leq \frac{p\sqrt{q}}{2^k} + \frac{q^{3/2}k\ell}{2^n} + 2\left(\frac{6p}{2^k}\right)^{2^{k-n}k\ell}. \tag{67}$$

IX. Bounding Bad5 | $\overline{\textsf{Bad3}}$. Recall that the event Bad5 holds, if there exists an user i such that one of its Σ values Σ_a^i collides with an another Σ values Σ_b^i and their corresponding tag collides. Since we bound the event Bad5 holds conditioned that $\overline{\mathsf{Bad3}}$ holds, it holds that the underlying block cipher key K_1^i has not collided with any chosen-ideal cipher key and hence, the V values are random. Therefore, by using the randomness of V values, and following Lemma 4, we bound the probability of $\Sigma_a^i = \Sigma_b^i$ to at most $2/2^n$. Moreover, the event $T_a^i = T_b^i$ is independent over $\Sigma_a^i = \Sigma_b^i$, which additionally contributes to the probability a factor 2^{-n}. Hence, by varying over all the possible choices of indices, we have

$$\Pr[\mathsf{Bad5} \mid \overline{\mathsf{Bad3}}] \leq \frac{q.q_{\max}}{2^{2n}}. \tag{68}$$

X. Bounding Bad6 | $\overline{\textsf{Bad4}}$. Bounding the event Bad6 | $\overline{\mathsf{Bad4}}$, is exactly identical to that of Bad5 | $\overline{\mathsf{Bad3}}$ and hence, following the result of Lemma 8, we have

$$\Pr[\mathsf{Bad6} \mid \overline{\mathsf{Bad4}}] \leq \frac{q.q_{\max}}{2^{2n}}. \tag{69}$$

XI. Bounding Bad7 | $\overline{\textsf{BadK4}}$. Recall that the event Bad7 holds, if there exists a pair of users i_1, i_2 such that one of the Σ values of i_1 user $\Sigma_a^{i_1}$ collides with an another Σ values of i_2 user $\Sigma_b^{i_2}$ and their corresponding key, i.e., $K_2^{i_1}$ collides with $K_2^{i_2}$. Since we bound the event Bad7 holds conditioned that $\overline{\mathsf{BadK4}}$ holds, it holds that at least one of the block cipher keys $K_1^{i_1}$ or $K_1^{i_2}$ has not collided with any chosen-ideal cipher key and hence, the corresponding V values are random. Without loss of generality, we assume that the V values of the i_2 user are random. Therefore, by using the randomness of V values of i_2 user, and following Lemma 4, we bound the probability of $\Sigma_a^{i_1} = \Sigma_b^{i_2}$ to at most $2/2^n$. Moreover, the event $K_2^{i_1} = K_2^{i_2}$ is independent over $\Sigma_a^{i_1} = \Sigma_b^{i_2}$, which additionally contributes to the probability a factor 2^{-k}. Hence, by varying over all the possible choices of indices, we have

$$\Pr[\mathsf{Bad7} \mid \overline{\mathsf{BadK4}}] \leq \frac{q^2}{2^{n+k}}. \tag{70}$$

XII. Bounding Bad8 | $\overline{\textsf{BadK4}}$. Bounding the event Bad8 | $\overline{\mathsf{BadK4}}$ is exactly identical to that of Bad7 | $\overline{\mathsf{BadK4}}$ and hence following the above analysis, we have

$$\Pr[\mathsf{Bad8} \mid \overline{\mathsf{BadK4}}] \leq \frac{q^2}{2^{n+k}}. \tag{71}$$

XIII. Bounding Bad9 | $\overline{\text{BadK4}}$. Recall that the event Bad9 holds, if there exists a pair of users i_1, i_2 such that one of the Θ values of i_1 user $\Theta_a^{i_1}$ collides with an another Θ values of i_2 user $\Theta_b^{i_2}$ and their corresponding key, i.e., $K_3^{i_1}$ collides with $K_3^{i_2}$. Since we bound the event Bad9 holds conditioned that $\overline{\text{BadK4}}$ holds, it holds that at least one of the block cipher keys $K_1^{i_1}$ or $K_1^{i_2}$ has not collided with any chosen-ideal cipher key and hence, the corresponding V values are random. Without loss of generality, we assume that the V values of i_2 user are random. Therefore, by using the randomness of V values of i_2 user, and following Lemma 8, we bound the probability of $\Theta_a^{i_1} = \Theta_b^{i_2}$ to at most $2/2^n$. Moreover, the event $K_3^{i_1} = K_3^{i_2}$ is independent over $\Theta_a^{i_1} = \Theta_b^{i_2}$, which additionally contributes to the probability a factor 2^{-k}. Hence, by varying over all the possible choices of indices, we have

$$\Pr[\mathsf{Bad9} \mid \overline{\mathsf{BadK4}}] \leq \frac{q^2}{2^{n+k}}. \tag{72}$$

XIV. Bounding Bad10 | $\overline{\text{BadK4}}$. Bounding the event Bad10 | $\overline{\text{BadK4}}$ is exactly identical to that of Bad9 | $\overline{\text{BadK4}}$ and hence following the above analysis, we have

$$\Pr[\mathsf{Bad10} \mid \overline{\mathsf{BadK4}}] \leq \frac{q^2}{2^{n+k}}. \tag{73}$$

XV. Bounding Bad11 | $\overline{\text{Bad3}} \wedge \overline{\text{Bad4}}$. Recall that the event Bad11 holds, if there exists an user i such that one of its (Σ, Θ) values (Σ_a^i, Θ_a^i) collides with an another (Σ, Θ) values (Σ_b^i, Θ_b^i). Since we bound the event Bad11 holds conditioned that $\overline{\text{Bad3}}$ and $\overline{\text{Bad4}}$ hold, it holds that the block cipher key K_1^i has not collided with any chosen-ideal cipher key and hence, the corresponding V values are random. Therefore, by using the randomness of V values, and following Lemma 7, we bound the probability of $(\Sigma_a^i = \Sigma_b^i) \wedge (\Theta_a^i = \Theta_b^i)$ to at most $4/2^{2n}$. Hence, by varying over all the possible choices of indices, we have

$$\Pr[\mathsf{Bad11} \mid \overline{\mathsf{Bad3}} \wedge \overline{\mathsf{Bad4}}] \leq \frac{2q.q_{\max}}{2^{2n}}. \tag{74}$$

XVI. Bounding Bad12 | $\overline{\text{Bad3}} \wedge \overline{\text{Bad4}}$. Recall that the event Bad12 holds, if there exists an user i such that $\Sigma_a^i = \Sigma_b^i$ and $\Theta_a^i = \Theta_c^i$ hold, where $a, b, c \in [q_i]$. Since we bound the event Bad12 holds conditioned that $\overline{\text{Bad3}}$ and $\overline{\text{Bad4}}$ hold, it holds that the block cipher key K_1^i has not collided with any chosen-ideal cipher key and hence, the corresponding V values are random. Therefore, by using the randomness of V values, and following Lemma 7, we bound the probability of $(\Sigma_a^i = \Sigma_b^i) \wedge (\Theta_a^i = \Theta_c^i)$ to at most $4/2^{2n}$. Hence, by varying over all the possible choices of indices, we have

$$\Pr[\mathsf{Bad12} \mid \overline{\mathsf{Bad3}} \wedge \overline{\mathsf{Bad4}}] \leq \frac{2q.q_{\max}^2}{3.2^{2n}}. \tag{75}$$

XVII. Bounding Bad13 and Bad14: For a fixed choice of indices, we define an indicator random variable $\mathbb{I}_{a,b}^i$ which takes the value 1, if $\Sigma_a^i = \Sigma_b^i$, and 0, otherwise. Let $\mathbb{I}^i = \sum_{a \neq b} \mathbb{I}_{a,b}^i$. By linearity of expectation,

$$\mathbf{E}[\mathbb{I}^i] = \sum_{a \neq b} \mathbf{E}[\mathbb{I}_{a,b}^i] = \sum_{a \neq b} \Pr[\Sigma_a^i = \Sigma_b^i] \overset{(1)}{\leq} \frac{q_i^2}{2^n},$$

where (1) holds from Lemma 4. Now,

$$\begin{aligned}
\Pr[\mathsf{Bad13}] &\leq \sum_{i \in [u]} \Pr[|\{(a,b) \in [q_i]^2 : \Sigma_a^i = \Sigma_b^i\}| \geq q_i^{1/2}] \\
&= \sum_{i=1}^{u} \Pr[\mathbb{I}^i \geq q_i^{1/2}] \overset{(2)}{\leq} \sum_{i=1}^{u} \frac{q_i^2}{q_i^{1/2} 2^n} = \sum_{i=1}^{u} \frac{q^{3/2}}{2^n} \leq \frac{q\sqrt{q_{\max}}}{2^n},
\end{aligned} \tag{76}$$

where (2) holds due to the Markov inequality. Similar to $\mathsf{Bad13}$, we bound $\mathsf{Bad14}$ as follows:

$$\Pr[\mathsf{Bad14}] \leq \frac{q\sqrt{q_{\max}}}{2^n}, \tag{77}$$

where the inequality follows from the result of Lemma 8. Finally, by combining Eqn. (55)-Eqn. (77), and the trivial inequality that

$$\left(\frac{6p}{2^k}\right)^{2^{k-n}k\ell} \leq \frac{6p}{2^k},$$

we obtain the bound of Lemma 9.